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ANTIBODY SPECIFICALLY BINDING TO CD47 AND ANTIGEN-BINDING FRAGMENT THEREOF

Abstract

Provided are an antibody specifically binding to CD47 and an antigen-binding fragment thereof. The antibody and the antigen-binding fragment thereof have high specificity and stability, can specifically block a CD47-SIPR α pathway, do not cause hemagglutination within a certain concentration range, and thus can significantly suppress tumor growth in vivo.

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Background/Summary

FIELD OF THE INVENTION

[0001] The invention relates to the research field of an antibody and the humanization transformation of the antibody, and in particular, the invention relates to an antibody capable of specifically binding to CD47 and an antigen-binding fragment thereof.

BACKGROUND OF THE INVENTION

[0002] CD47, also known as integrin-associated protein (IAP), belongs to the immunoglobulin superfamily, and its structure mainly includes an extracellular immunoglobulin variable region-like domain, a highly hydrophobic five-transmembrane domain, and an intracellular carboxy-terminal cytoplasmic tail domain selectively spliced. CD47 is widely expressed on the cell surface and can bind to the signal regulatory protein SIPR α , integrin and thrombospondin TSP1, and is involved in a variety of physiological functions. The main function thereof is to inhibit the phagocytosis of macrophages by binding to SIPR α on the surface of phagocytes. SIPR α is also a member of the immunoglobulin superfamily, and its intracellular domain contains a tyrosine-rich immunoreceptor tyrosine-based inhibitory motif (ITIM). When CD47 binds to SIPR α on the surface of macrophages, ITIM is phosphorylated and binds to the downstream molecule SHP1/SHP2, preventing the accumulation of myosin in the phagocytic synapse submembrane to inhibit phagocytosis. Under normal conditions, CD47 is a “self” signal representing “don't eat me”, which can block the autoaggression of phagocytosis; under pathological conditions, CD47 is highly expressed on tumor cells, helping tumor cells escape the monitoring of phagocytes (Science, 2000, 288(5473): 2051-2054; Trends in Cell Biology, 2001, 11(3):130-135).

[0003] Tumor studies have found that CD47 is expressed in a variety of human tumor types, including acute myeloid leukemia, chronic myeloid leukemia, acute lymphoblastic leukemia, non-Hodgkin's lymphoma, multiple myeloma, bladder cancer, and other solid tumors. Tumor cells can escape the immune surveillance of macrophages by up-regulating highly expressed CD47 and combining with SIPR α . Therefore, blocking the interaction between CD47 and SIPR α (with a CD47 antibody or SIPR α -Fc) can activate the phagocytosis of a tumor by macrophages and exert a specific killing effect of CD8⁺ T cells on the tumor through the presentation of antigens to CD8⁺ T cells by DC cells (Current Opinion in Immunology, 2012, 24(2): 225-232; Nature Medicine, 2015, 21(10): 1209).

[0004] Antibodies that block the CD47-SIPR α pathway have broad application prospects, because almost all tumor cell types express CD47, meaning a broad spectrum in anti-tumor; CD47 serves as an immune checkpoint, and some studies have found that the combinations of CD47 antibodies with the medicaments against other immune checkpoints CTLA4/PD-1 can enhance the potency. However, since red blood cells also express CD47, the safety of anti-CD47 antibodies needs to be considered. Studies have found that in addition to blocking the “don't eat me” signal of CD47, phagocytosis exerted by macrophages also transmits a “eat me” signal mediated by calreticulin (CRT). Normally, tumor cells highly express calreticulin, while normal cells do not express it. In addition, when screening CD47 antibodies, antibodies that do not cause erythrocytes agglutination and which are in the form of IgG4 are selected, which will not activate the phagocytosis of red blood cells by macrophages. There remains a need in the art for the anti-CD47 antibodies.

SUMMARY OF THE INVENTION

[0005] In one aspect, the invention relates to an isolated anti-human CD47 antibody, an antigen-

binding fragment, or a variant thereof, wherein said antibody or antigen-binding fragment thereof comprises a light chain variable region and/or a heavy chain variable region, wherein [0006] the amino acid sequences of LCDR1, LCDR2, and LCDR3 of the light chain variable region are set forth in SEQ ID Nos. 1, 2, and 3, respectively; and/or [0007] the amino acid sequence of HCDR1 of the heavy chain variable region is set forth in SEQ ID No. 4; the amino acid sequence of HCDR2 of the heavy chain variable region has at least about 94% identity to SEQ ID No. 5 or 6; the amino acid sequence of HCDR3 of the heavy chain variable region has at least about 78% identity to SEQ ID No. 7 or 8; or [0008] the amino acid sequences of the CDRs of the light chain variable region or the heavy chain variable region have at least 70% identity to the sequences set forth in SEQ ID Nos. 1-8 respectively and retain the biological activity of the corresponding parent sequences, such as at least about 75%, about 80%, about 85%, about 90% or about 95% or higher identity; or [0009] the amino acid sequences of the CDRs of the light chain variable region or the heavy chain variable region are variant sequences of the sequences set forth in SEQ ID Nos. 1-8 respectively obtained after deletion, substitution and/or addition of one or more amino acid residues which retain the biological activities of the corresponding parent sequences, such as 1, 2, 3 or 4 or more amino acid residues; [0010] wherein the variant is one selected from the group consisting of chimeric antibodies, humanized antibodies, and fully human antibodies.

[0011] In some embodiments, the heavy chain constant region sequence of the antibody is the constant region sequence of one selected from the group consisting of human IgG1, IgG2, IgG3, IgG4, IgA, IgM, IgE, IgD, and/or, the light chain constant region sequence of the antibody is the constant region sequence of κ chain or λ chain; preferably, the heavy chain constant region sequence is the constant region sequence of IgG1 or IgG4, and/or the light chain constant region sequence is the constant region sequence of κ light chain.

[0012] In some embodiments, the amino acid sequence of the light chain variable region of a chimeric anti-CD47 antibody or a functional fragment thereof is as set forth in SEQ ID NO. 9 or 10; or has at least 70% identity to the sequence set forth in SEQ ID No. 9 or 10 and retains the biological activity of the corresponding parent sequence, such as at least about 75%, about 80%, about 85%, about 90%, about 95%, about 96%, about 97%, about 98%, about 99% or higher identity; or is a variant sequence of the sequence set forth in SEQ ID NO. 9 or 10 obtained by deleting, substituting and/or adding one or more amino acid residues which retains the biological activity of the corresponding parent sequence, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15 or more amino acid residues; and/or the amino acid sequence of the heavy chain variable region is as set forth in SEQ ID NO. 11 or 12; or has at least 70% identity to the sequence set forth in SEQ ID NO. 11 or 12 and retains the biological activity of corresponding parent sequence, such as at least about 75%, about 80%, about 85%, about 90%, about 95%, about 96%, about 97%, about 98%, about 99% or higher identity; or is a variant sequence of the sequence set forth in SEQ ID NO. 11 or 12 obtained by deleting, substituting and/or adding one or more amino acid residues that retains the biological activity of the corresponding parent sequence, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15 or more amino acid residues; and/or [0013] the amino acid sequences of the light chain constant region and the heavy chain constant region of the chimeric anti-CD47 antibody or a functional fragment thereof are set forth in SEQ ID NO. 13 and SEQ ID Nos. 14-16 respectively; or have at least 70% identity to the sequences set forth in SEQ ID NO. 13 and SEQ ID Nos. 14-16 respectively and retain the biological activity of the corresponding parent sequences, such as at least about 75%, about 80%, about 85%, about 90%, about 95%, about 96%, about 97%, about 98%, about 99%, about 99.2%, about 99.5% or higher identity; or are variant sequences of the sequences set forth in SEQ ID NO. 13 and SEQ ID Nos. 14-16 respectively obtained by deleting, substituting and/or adding one or more amino acid residues that retain the biological activity of the corresponding parent sequences, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15 or more amino acid residues.

[0014] In some embodiments, the framework regions of the light chain variable region of the anti-human CD47 antibody, antigen-binding fragment or variant thereof include FR-L1, FR-L2, FR-L3

and FR-L4, and the framework regions of the heavy chain variable include FR-H1, FR-H2, FR-H3 and FR-H4, and/or wherein, [0015] the amino acid sequence of the FR-L1 is set forth in SEQ ID NO. 17; [0016] the amino acid sequence of the FR-L2 is set forth in SEQ ID NO. 18, or an amino acid sequence as set forth in SEQ ID NO. 18 obtained by one of the following substitutions or any combination thereof: [0017] the 12th amino acid R is substituted with T; [0018] the amino acid sequence of the FR-L3 is set forth in SEQ ID NO. 19, or an amino acid sequence as set forth in SEQ ID NO. 19 obtained by one of the following substitutions or any combination thereof: [0019] the 31st amino acid Y is substituted with F; [0020] the amino acid sequence of the FR-L4 is set forth in SEQ ID NO. 20; [0021] the amino acid sequence of the FR-H1 is set forth in SEQ ID NO. 21; [0022] the amino acid sequence of the FR-H2 is set forth in SEQ ID NO. 22; [0023] the amino acid sequence of the FR-H3 is set forth in SEQ ID NO. 23, or an amino acid sequence as set forth in SEQ ID NO. 23 obtained by one of the following substitutions or any combination thereof: [0024] the 8th amino acid E is substituted with T, [0025] the 11th amino acid S is substituted with N, [0026] the 31st amino acid A is substituted with V; and/or [0027] the amino acid sequence of the FR-H4 is set forth in SEQ ID NO. 24.

[0028] In some embodiments, the amino acid sequence of the light chain variable region is as set forth in any one of SEQ ID Nos. 25-27, and/or the amino acid sequence of the heavy chain variable region is as set forth in any one of SEQ ID Nos. 28-33; or is a variant sequence having at least 70% identity to one of the amino acid sequences set forth SEQ ID Nos. 25-27 or 28-33 and retaining the biological activity of the corresponding parent sequence, such as at least about 75%, about 80%, about 85%, about 90%, about 95%, about 96%, about 97%, about 98%, about 99% or higher identity; or is a variant sequence of one of the sequences set forth in SEQ ID Nos. 25-27 or 28-33 obtained after deletion, substitution and/or addition of one or more amino acid residues which retains the biological activity of the corresponding parent sequence, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15 or more amino acid residues; [0029] preferably, the light chain variable region sequence is as set forth in SEQ ID NO. 26 or 27, and/or the heavy chain variable region sequence is as set forth in SEQ ID NO. 33.

[0030] It is well known in the art that when referring to an identity, since the length of an amino acid sequence can only be a natural number, the actual identity value obtained by calculation may not be a finite percentage like 95%, but a number close to a percentage such as 95%. For example, for the variable region sequence of 117 amino acid residues, when only one amino acid residue is changed, the corresponding percentage of identity is a percentage close to 99.15% actually, but for convenience, such a figure would only be recorded as 99% in the context.

[0031] In some embodiments, the antigen-binding fragment is one or more selected from the group consisting of F(ab').sub.2, Fab', Fab, Fd, Fv, scFv, diabody, camelid antibody, CDR, and antibody minimal recognition unit (dAb), preferably, the antigen-binding fragment is Fab, F(ab').sub.2 or scFv.

[0032] Another aspect of the invention relates to an isolated nucleic acid molecule selected from the group consisting of: [0033] (1) DNA or RNA encoding the anti-human CD47 antibody, antigen-binding fragment or variant thereof as described above; [0034] (2) a nucleic acid that is completely complementary to the nucleic acid defined in (1).

[0035] Yet another aspect of the invention relates to a vector comprising the above-mentioned nucleic acid molecule operably linked, preferably, the vector is an expression vector.

[0036] One aspect of the invention relates to a host cell comprising the nucleic acid molecule or the vector as described above.

[0037] Another aspect of the invention relates to a composition comprising the anti-human CD47 antibody, antigen-binding fragment or variant thereof as described above, the nucleic acid molecule as described above, the vector as described above or the host cell as described above, and a pharmaceutically acceptable carrier, diluent, or excipient.

[0038] Another aspect of the invention relates to a method of producing the above-mentioned anti-

human CD47 antibody, antigen-binding fragment or variant thereof, comprising the steps of: [0039] culturing the above-mentioned host cells under culture conditions suitable for the expression of the anti-human CD47 antibody, antigen-binding fragment or variant thereof, and optionally, isolating and purifying the obtained products.

[0040] Another aspect of the invention relates to use of the anti-human CD47 antibody, antigen-binding fragment or variant thereof as described above, the nucleic acid molecule as described above, the vector as described above or the host cell as described above in the manufacture of a medicament for the prevention and/or the treatment of CD47-mediated diseases or disorders such as autoimmune diseases, immune responses against grafts, allergies, infectious diseases, neurodegenerative diseases and tumors.

[0041] Another aspect of the invention relates to the anti-human CD47 antibody, antigen-binding fragment or variant thereof as described above, the nucleic acid molecule as described above, the vector as described above or the host cell as described above, for use in the prevention and/or treatment of CD47-mediated diseases or disorders such as autoimmune diseases, immune responses to grafts, allergies, infectious diseases, neurodegenerative diseases, and tumors.

[0042] Another aspect of the invention relates to a method of preventing and/or treating CD47-mediated diseases or disorders such as autoimmune diseases, immune responses to grafts, allergies, infectious diseases, neurodegenerative diseases and tumors, comprising administering to a subject in need thereof the anti-human CD47 antibody, antigen-binding fragment or variant thereof as described above, the nucleic acid molecule as described above, the vector as described above or the host cell as described above.

[0043] In some embodiments, the autoimmune diseases are one or more selected from the group consisting of arthritis, rheumatoid arthritis, psoriasis, multiple sclerosis, ulcerative colitis, Crohn's disease, systemic lupus erythematosus, glomerulonephritis, dilated cardiomyopathy-like disease, Sjogren's syndrome, allergic contact dermatitis, polymyositis, scleroderma, periarteritis polyarteritis, rheumatic fever, vitiligo, insulin-dependent diabetes mellitus, Behcet's syndrome and chronic thyroiditis; preferably, the autoimmune diseases are one or more selected from the group consisting of arthritis, rheumatoid arthritis, psoriasis, multiple sclerosis, ulcerative colitis, Crohn's disease, systemic lupus erythematosus, glomerulonephritis, rheumatic fever, vitiligo, insulin-dependent diabetes and chronic thyroiditis; more preferably, the autoimmune diseases are one or more selected from the group consisting of rheumatoid arthritis, psoriasis, multiple sclerosis, ulcerative colitis, Crohn's disease, systemic lupus erythematosus, insulin-dependent diabetes mellitus, and chronic thyroiditis.

[0044] In some embodiments, the immune response against the graft includes, for example, graft versus host disease.

[0045] In some embodiments, the allergies are one or more selected from the group consisting of urticaria, eczema, angioedema, allergic rhinitis, bronchial asthma, laryngeal edema, food allergic gastroenteritis, and anaphylactic shock; preferably, the allergies are one or more selected from the group consisting of urticaria, eczema, allergic rhinitis, bronchial asthma, and anaphylactic shock; more preferably, the allergies are one or more selected from the group consisting of allergic rhinitis, bronchial asthma, and anaphylactic shock.

[0046] In some embodiments, the infectious disease refers to the local and systemic inflammatory response caused by the invasion of viruses, bacteria, fungi, parasites and other pathogens and specific toxins into the human body. Examples of pathogenic viruses include, for example, HIV, hepatitis viruses (types A, B, and C), herpes viruses (e.g., VZV, HSV-1, HAV-6, HSV-II, and CMV, Epstein-Barr virus), adenoviruses, influenza virus, flavivirus, echovirus, rhinovirus, coxsackievirus, coronavirus, respiratory syncytial virus, mumps virus, rotavirus, measles virus, rubella virus, parvovirus, vaccinia virus, HTLV virus, Dengue virus, papilloma virus, molluscum virus, polio virus, rabies virus, JC virus, and arboencephalitis virus. Pathogenic bacteria include, for example, *Treponema pallidum*, *Chlamydia*, *Rickettsia*, *Mycobacteria*, *Staphylococci*, *Streptococci*,

Pneumococci, *Meningococci* and *Gonorrhoeae* (*conococci*), *Klebsiella*, *Proteus*, *Serratia*, *Pseudomonas*, *Legionella*, *Diphtheria bacillus*, *Salmonella*, *Bacillus*, *Cholera bacteria*, *Clostridium tetani*, *Bacillus botulinus*, *Bacillus anthracis*, *Yersinia pestis*, *Leptospirosis bacteria* and *Borrelia burgdorferi* causing Lyme disease. Pathogenic fungi include, for example, *Candida* (*Candida albicans*, *Candida krusei*, *Candida glabrata*, *Candida tropicalis*, etc.), *Cryptococcus neoformans*, *Aspergillus* (*Aspergillus fumigatus*, *Aspergillus niger*, etc.), *Mucorales* (*mucor*, *bsidia*, *rhizophus*)), *Sporothrix schenckii*, *Blastomyces dermatitidis*, *Paracoccidioides brasiliensis*, *Coccidioides immitis*, and *Histoplasma capsulatum*. Pathogenic parasites include, for example, *Entamoeba histolytica*, *Balantidium coli*, *Naegleria fowleri*, *Acanthamoeba* sp., *Giardialambia*, *Cryptosporidium* sp., *Pneumocystis carinii*, *Plasmodium vivax*, *Babesia microti*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Leishmania donovani*, *Toxoplasma gondi* and *Nippostrongylus brasiliensis*.

[0047] In some embodiments, the neurodegenerative diseases are one or more selected from the group consisting of Parkinson's disease, Huntington's disease, Machado-Joseph disease, amyotrophic lateral sclerosis, Creutzfeldt-Jakob disease. Preferably, the neurodegenerative diseases are one or more selected from the group consisting of Parkinson's disease, Huntington's disease, and amyotrophic lateral sclerosis; more preferably, the neurodegenerative diseases are one or more selected from consisting of Parkinson's disease and Huntington's disease.

[0048] In some embodiments, the tumors are one or more selected from the group consisting of leukemia, lymphoma, myeloma, brain tumor, head and neck squamous cell carcinoma, non-small cell lung cancer, small cell lung cancer, nasopharyngeal cancer, esophageal cancer, gastric cancer, pancreatic cancer, gallbladder cancer, liver cancer, colorectal cancer, breast cancer, ovarian cancer, cervical cancer, endometrial cancer, uterine sarcoma, prostate cancer, bladder cancer, urothelial cancer, renal cell carcinoma, osteosarcoma, melanoma, and merkel cell carcinoma; preferably, the tumors are one or more selected from the group consisting of lymphoma, myeloma, head and neck squamous cell carcinoma, non-small cell lung cancer, small cell lung cancer, nasopharyngeal cancer, esophageal cancer, gastric cancer, liver cancer, colorectal cancer, breast cancer, cervical cancer, endometrial cancer, prostate cancer, urothelial cancer, renal cell carcinoma, osteosarcoma, melanoma; more preferably, the tumors are one or more selected from the group consisting of lymphoma, head and neck squamous cell carcinoma, non-small cell lung cancer, gastric cancer, liver cancer, colorectal cancer, cervical cancer, urothelial carcinoma, renal cell carcinoma, and melanoma.

[0049] In some embodiments, the subject is selected from mammals, including but not limited to human and/or other primates; mammals, including commercially relevant mammals such as cattle, pigs, horses, sheep, cats, dogs, mice and/or rats.

[0050] In some embodiments, the anti-human CD47 antibody, antigen-binding fragment or variant thereof, the nucleic acid molecule, the vector or the host cell of the invention is utilized by conventional administration methods in the art, such as parenteral route, such as intravenous injection.

[0051] In other words, the invention obtains the anti-CD47 monoclonal antibody by means of hybridoma, and screens the antibodies that do not cause erythrocytes agglutination. By using the IgG4 antibody form, a humanized IgG4 antibody that can specifically block the CD47-SIPR α pathway and does not cause erythrocytes agglutination within a certain concentration range was obtained.

[0052] The antibodies and functional fragments thereof provided by the invention can specifically bind human CD47 with a K.sub.D of 1 nM or less, block the binding activity of CD47 to the receptors thereof, mediate phagocytosis of tumor cells by macrophages and/or does not cause erythrocytes agglutination, and can be used to prevent and/or treat autoimmune diseases, immune responses against grafts, allergies, infectious diseases, neurodegenerative diseases and tumors.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0053] To illustrate the embodiments of the invention or the technical solutions in the prior art more clearly, the following will briefly introduce the accompanying drawings that need to be used in the embodiments or the prior art. Apparently, the accompanying drawings in the following description show some embodiments of the invention, and for those skilled in the art, they can obtain other accompanying drawings based on these accompanying drawings without any creative work.

[0054] FIG. 1 shows the in vitro activity of the anti-human CD47 murine monoclonal antibody secreted by the anti-human CD47 positive clone in Example 1. FIG. 1A shows the binding activity of the murine monoclonal antibody to CD47, and FIG. 1B shows the binding activity of the murine monoclonal antibody to CD47/SIRP α .

[0055] FIG. 2 shows the effects of the anti-human CD47 murine monoclonal antibodies secreted by the anti-human CD47 positive clones in Example 1 on red blood cells.

[0056] FIG. 3 shows the binding activities of the chimeric anti-human CD47 monoclonal antibodies to human CD47 in Example 3.

[0057] FIG. 4 shows the specificities of the chimeric anti-human CD47 monoclonal antibodies in Example 4. FIG. 4A shows the species specificities, and FIG. 4B shows the binding specificities.

[0058] FIG. 5 shows the activities of the chimeric anti-human CD47 monoclonal antibodies in Example 5 in blocking the binding to CD47/SIRP α .

[0059] FIG. 6 shows the phagocytic activities on the tumor cells mediated by the chimeric anti-human CD47 monoclonal antibodies in Example 6.

[0060] FIG. 7 shows the effects of the chimeric anti-human CD47 monoclonal antibodies on red blood cells in Example 7.

[0061] FIG. 8 shows the in vivo anti-tumor efficacies of the chimeric anti-human CD47 monoclonal antibodies in Example 8.

[0062] FIG. 9 shows the antigen-binding activities of the anti-human CD47 humanized monoclonal antibodies in Example 10.

[0063] FIG. 10 shows the activities of the anti-human CD47 humanized monoclonal antibodies in Example 11 of blocking the binding to CD47/SIRP α .

[0064] FIG. 11 shows the effects of the anti-human CD47 humanized monoclonal antibodies in Example 13 on red blood cells.

[0065] FIG. 12 shows the in vivo anti-tumor efficacies of the anti-human CD47 humanized monoclonal antibodies in Example 16.

DETAILED DESCRIPTION OF THE EMBODIMENTS

Definitions

[0066] The term “CD47” refers to the integrin-associated protein (TAP), which may be a CD47 molecule of any mammalian origin. In some embodiments, it is a CD47 molecule of primate origin, and in some embodiments, it is a CD47 molecule of human origin.

[0067] As used herein, the terms “anti-CD47 antibody”, “anti-CD47”, “CD47 antibody” or “antibody binding to CD47” refer to an antibody which is capable of binding to the CD47 protein or a fragment thereof with sufficient affinity such that the antibody can be used as a diagnostic and/or therapeutic agent targeting CD47. In some embodiments, the anti-CD47 antibody binds to an epitope of CD47 that is conserved among CD47s from different species.

[0068] The term “antibody” refers to an immunoglobulin molecule or a fragment of an immunoglobulin molecule which can bind to an epitope of an antigen. Naturally occurring antibodies typically comprise a tetramer which is usually composed of at least two heavy (H) chains and at least two light (L) chains. Immunoglobulins include the following isotypes: IgG, IgA,

IgM, IgD, and IgE, with their corresponding heavy chains being the μ , δ , γ , μ , and ϵ chains, respectively. The Igs of the same type can be divided into different subtypes according to the amino acid compositions of the hinge regions and the number and positions of the heavy chain disulfide bonds. For example, IgG can be divided into IgG1, IgG2, IgG3, and IgG4 subtypes, and IgA can be divided into IgA.sub.1 and IgA.sub.2 subtypes. Light chains can be divided into κ and λ chains according to the constant regions.

[0069] Herein, the term “antibody” is used in the broadest sense thereof, and refers to a protein comprising an antigen-binding site, encompassing natural and artificial antibodies of various structures, including but not limited to intact antibodies and antigen-binding fragments of antibodies.

[0070] A “variable region” or “variable domain” is a domain of a heavy or light chain of an antibody involving in the binding of the antibody to its antigen. Each heavy chain of an antibody consists of a heavy chain variable region (herein referred to as VH) and a heavy chain constant region (herein referred to as CH), wherein the heavy chain constant region usually consists of 3 domains (CH1, CH2 and CH3). Each light chain consists of a light chain variable region (herein referred to as VL) and a light chain constant region (herein referred to as CL). The variable regions of the heavy and light chains are typically responsible for antigen recognition, while the constant domains of the heavy and light chains can mediate the binding of the immunoglobulin with host tissues or factors (including various cells of the immune system (e.g., effector cells), Fc receptors and the first component of the classical complement system (C1q)). The heavy and light chain variable regions comprise binding regions interacting with an antigen. The VH and VL regions can be further subdivided into hypervariable regions (HVRs) called “complementarity determining regions (CDRs)”, which are interspersed with more conserved regions called “framework regions (FRs)”. Each VH or VL consists of three CDR domains and four FR domains, arranged from the amino-terminus to the carboxy-terminus in the following order: FR1-CDR1-FR2-CDR2-FR3-CDR3-FR4.

[0071] The terms “complementarity determining region” or “CDR region” or “CDR” (used interchangeably herein with the hypervariable region “HVR”) refer to a region in the antibody variable domains which is hypervariable in sequence and forms a loop structurally determined (“a hypervariable loop”) and/or comprises antigen-contacting residues (“antigen-contacting points”). The CDRs are primarily responsible for binding to an epitope. Herein, the three CDRs of the heavy chain are referred to as HCDR1, HCDR2 and HCDR3, and the three CDRs of the light chain are referred to as LCDR1, LCDR2 and LCDR3.

[0072] It should be noted that the boundaries of the CDRs of the variable regions of the same antibody obtained based on different numbering systems may be different from each other. That is, the CDR sequences of the variable regions of the same antibody defined under different numbering systems are somewhat different from each other. Thus, when defining an antibody with the particular CDR sequences as defined in the invention, the scope of said antibody also covers antibodies of which the variable region sequences comprise said particular CDR sequences, but due to the different numbering system(s) used (such as different numbering system rules or combinations thereof), the CDR boundaries claimed by them might be different from the specific CDR boundaries defined in the invention.

[0073] The terms “mAbs”, “monoclonal antibodies” or “monoclonal antibody composition” refer to a preparation of antibody molecules of single molecular composition, and refer to antibodies obtained from a substantially homogeneous population of antibodies, i.e., the population comprising individual antibodies are identical except for the naturally occurring mutations that may be present in minor amounts. A conventional monoclonal antibody compositions displays a single binding specificity and affinity for a particular epitope. In certain embodiments, a monoclonal antibody can be composed of more than one Fab domain thereby increasing the specificity to more than one target. The term “monoclonal antibody” or “monoclonal antibody composition” is not

limited by any particular method of production (eg, recombinant, transgenic, hybridoma, etc.).

[0074] The terms “double antibody”, “bifunctional antibody”, “bispecific antibody” or “BsAb” refers to an antibody which has two different antigen binding sites, so that which can simultaneously bind to two target antigens, alternatively which plays a role in mediating another special function simultaneously besides the targeting effect of the antibody. The special function effector molecules mediated can also be toxins, enzymes, cytokines, radionuclides, etc. The two arms of the bispecific antibody binding to an antigen can be from Fab, Fv, ScFv or dSFv, etc.

[0075] The term “polyclonal antibody” refers to a preparation comprising different antibodies raised against different determinants (“epitopes”).

[0076] The term “antigen-binding fragment of antibody” refers to a fragment, a part, a region or a domain of an antibody capable of binding to an epitope (e.g., obtainable by cleavage, recombination, synthesis, etc.). An antigen-binding fragment may comprise 1, 2, 3, 4, 5 or all 6 CDR domains of such antibodies and, while capable of binding to the epitope, it may exhibit different specificities, affinities or selectivities. Preferably, an antigen-binding fragment comprises all 6 CDR domains of said antibody. An antigen-binding fragment of an antibody may be a part of a single polypeptide chain (e.g., scFv) or comprise a single polypeptide chain, or may be a part of two or more polypeptide chains (each having an amino- and carboxy-terminus) (e.g., a bispecific antibody, a Fab fragment, a F(ab')₂ fragment, etc.) or comprise two or more polypeptide chains.

[0077] Examples of an antigen-binding fragment of the invention include (a) a Fab' or Fab fragment, a monovalent fragment consisting of the VL, VH, CL and CH1 domains; (b) F(ab')₂ fragments, bivalent fragments comprising two Fab fragments linked by a disulfide bond at the hinge domain; (c) an Fd fragment consisting of the VH and CH1 domains; (d) a Fv fragment consisting of VL and VH domains of a single arm of an antibody; (e) a single chain Fv (scFv), a recombinant protein formed by linking VH and VL domains of an antibody with a peptide linker by genetic engineering; (f) a dAb fragment (Ward et al., *Nature*, 341, 544-546 (1989)), which consists essentially of a VH domain and also called domain antibodies (Holt et al., *Trends Biotechnol.*, 2i(11): 484-90); (g) camelid or nanobodies (Reverts et al., *Expert Opin Biol Ther.*, 5(1): 111-24) and (h) an isolated complementarity determining region (CDR).

[0078] The term “chimeric antibody” means that a portion of the heavy and/or light chains is identical or homologous to the corresponding sequences of an antibody from a particular species or belonging to a particular antibody class or subclass, while the rest of the chains is identical or homologous to the corresponding sequences of an antibody from another species or belonging to another antibody class or subclass, providing that they exhibit the desired biological activity. The invention provides the variable region antigen binding sequences from human antibodies. Thus, chimeric antibodies primarily focused herein include antibodies which comprise one or more human antigen-binding sequences (e.g., CDRs) and comprise one or more sequences from non-human antibodies, such as FR or C region sequences. Furthermore, chimeric antibodies described herein are antibodies comprising human variable region antigen-binding sequence from one antibody class or subclass and other sequences from another antibody class or subclass, such as FR or C region sequences.

[0079] The term “humanized antibody” refers to an antibody in which CDR sequences from the germline of another mammalian species, such as a mouse, have been grafted among the human framework region sequences. Additional framework region modifications can be made within the human framework region sequences.

[0080] The term “human antibody” or “fully human antibody” (“humAb” or “HuMab”) refers to an antibody comprising variable and constant domains derived from human germline immunoglobulin sequences. The human antibodies of the invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or during gene rearrangement or by somatic mutation in vivo).

[0081] Variant antibodies are also included within the scope of the invention. Accordingly, variants of the sequences recited in the invention are also included within the scope of the invention. Other variants of antibody sequences with improved affinity can be obtained by using methods known in the art and these variants are also included within the scope of the invention. For example, amino acid substitutions can be used to obtain antibodies with further improved affinity. Alternatively, codon optimization of nucleotide sequences can be used to improve the translation efficiency of expression systems in antibody production.

[0082] Such variant antibody sequences have 70% or more (e.g., 80%, 85%, 90%, 95%, 97%, 98%, 99% or more) sequence homology to the sequences recited in the invention. Such a sequence homology is obtained by calculation relative to the full length of the reference sequence (i.e., the sequence recited in the invention).

[0083] The numbering of amino acid residues in the invention is according to IMGT®, the international ImmunoGeneTics information System® or Kabat, E. A., Wu, T. T., Perry, H. M., Gottesmann, K. S. & Foeller, C. (1991). Sequences of Proteins of Immunological Interest, 5th edit., NIH Publication no. 91-3242 U.S. Department of Health and Human Services; Chothia, C. & Lesk, A. M. (1987). Canonical structures For The Hypervariable domains Of Immunoglobulins. J. Mol. Biol. 196, 901-917. Unless otherwise specified, the numbering of amino acid residues in the invention is according to the Kabat numbering system.

[0084] An antibody or an antigen-binding fragment thereof is said to “specifically” bind a region of another molecule (i.e., an epitope) if it reacts or associates more frequently, more rapidly, with greater duration and/or with greater affinity or avidity with that epitope relative to alternative epitopes. In some embodiments, the antibody or antigen-binding fragment thereof of the invention binds human CD47 with an affinity of at least 10^{-7} M, such as 10^{-8} M, 10^{-9} M, 10^{-10} M, 10^{-11} M or higher. Preferably, the antibody or antigen-binding fragment thereof binds under physiological conditions (e.g., in vivo). Therefore, “specifically binding to CD47” refers to the ability of the antibody or antigen-binding fragment thereof to bind to CD47 with the above-mentioned specificity and/or under such conditions. Methods suitable for determining such a binding are known in the art.

[0085] The term “binding” in the context of the binding of an antibody to a predetermined antigen typically refers to binding with an affinity corresponding to a K_D of about 10^{-6} M or less, which is at least ten-fold lower, such as at least 100-fold lower, for instance at least 1,000-fold lower than its affinity for binding to a non-specific antigen (e.g., BSA, casein) other than the predetermined antigen or a closely-related antigen.

[0086] As used herein, the term “ k_d ” (sec-1 or 1/s) refers to the dissociation rate constant of a particular antibody-antigen interaction. Said value is also referred as the $k_{sub.off}$ value.

[0087] As used herein, the term “ k_a ” ($M^{-1} \times sec^{-1}$ or 1/Msec) refers to the association rate constant of a particular antibody-antigen interaction.

[0088] As used herein, the term “ K_D ” (M) refers to the dissociation equilibrium constant of a particular antibody-antigen interaction and is obtained by dividing the k_d by the k_a .

[0089] As used herein, the term “ K_A ” (M^{-1} or 1/M) refers to the association equilibrium constant of a particular antibody-antigen interaction and is obtained by dividing the k_a by the k_d .

[0090] In some embodiments, only part of a CDR, namely the subset of CDR residues required for binding, termed the SDRs, are needed to retain binding in a humanized antibody. CDR residues not contacting the relevant epitope and not in the SDRs can be identified based on previous studies (for example residues H60-H65 in CDR H2 are often not required), from regions of Kabat CDRs lying outside Chothia hypervariable loops (see, Kabat et al. (1992), “Sequences of Proteins of Immunological Interest”, National Institutes of Health Publication No. 91-3242; Chothia, C. et al. (1987), “Canonical Structures For The Hypervariable Regions Of Immunoglobulins”, J. Mol. Biol. 196:901-917), by molecular modeling and/or empirically, or as described in Gonzales, N. R. et al. (2004), “SDR Grafting Of A Murine Antibody Using Multiple Human Germline Templates To

Minimize Its Immunogenicity", Mol. Immunol. 41:863-872. In such humanized antibodies at positions in which one or more donor CDR residues is absent or in which an entire donor CDR is omitted, the amino acid occupying the position can be an amino acid occupying the corresponding position (by Kabat numbering) in the acceptor antibody sequence. The number of such substitutions of acceptor for donor amino acids in the CDRs to include reflects a balance of competing considerations. Such substitutions are potentially advantageous in decreasing the number of mouse amino acids in a humanized antibody and consequently decreasing potential immunogenicity. However, substitutions can also cause changes of affinity, and significant reductions in affinity are preferably avoided. Positions for substitution within CDRs and amino acids to substitute can also be selected empirically.

[0091] The fact that a single amino acid alteration of a CDR residue can result in loss of functional binding (Rudikoff, S. et al. (1982), "Single Amino Acid Substitution Altering Antigen-Binding Specificity", Proc. Natl. Acad. Sci. (USA) 79(6):1979-1983) provides a means for systematically identifying alternative functional CDR sequences. In one preferred method for obtaining such variant CDRs, a polynucleotide encoding the CDR is mutagenized (for example via random mutagenesis or by a site-directed method (e.g., polymerase chain-mediated amplification with primers that encode the mutated locus)) to produce a CDR having a substituted amino acid residue. By comparing the identity of the relevant residue in the original (functional) CDR sequence to the identity of the substituted (non-functional) variant CDR sequence, the BLOSUM62.ijj substitution score for that substitution can be identified. The BLOSUM system provides a matrix of amino acid substitutions created by analyzing a database of sequences for trusted alignments (Eddy, S. R. (2004), "Where Did The BLOSUM62 Alignment Score Matrix Come From?", Nature Biotech. 22(8):1035-1036; Henikoff, J. G. (1992), "Amino acid substitution matrices from protein blocks", Proc. Natl. Acad. Sci. (USA) 89:10915-10919; Karlin, S. et al. (1990), "Methods For Assessing The Statistical Significance Of Molecular Sequence Features By Using General Scoring Schemes", Proc. Natl. Acad. Sci. (USA) 87:2264-2268; Altschul, S. F. (1991), "Amino Acid Substitution Matrices From An Information Theoretic Perspective", J. Mol. Biol. 219, 555-565. Currently, the most advanced BLOSUM database is the BLOSUM62 database (BLOSUM62.ijj). Table 1 presents the BLOSUM62.ijj substitution scores (the higher the score the more conservative the substitution and thus the more likely the substitution will not affect function). If an epitope-binding fragment comprising the resultant CDR fails to bind to CD47, for example, then the BLOSUM62.ijj substitution score is deemed to be insufficiently conservative, and a new candidate substitution is selected and produced having a higher substitution score. Thus, for example, if the original residue was glutamate (E), and the non-functional substitute residue was histidine (H), then the BLOSUM62.ijj substitution score will be 0, and more conservative changes (such as to aspartate, asparagine, glutamine, or lysine) are preferred.

TABLE-US-00001																														TABLE 1																														A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V	A	+4	-1	-2	-2	0	-1	-1
0	-2	-1	-1	-1	-1	-2	-1	+1	0	-3	-2	0	R	-1	+5	0	-2	-3	+1	0	-2	0	-3	-2	+2	-1	-3	-2	-1	-1	-3	-2	-3	N	-2	0	+6	+1	-3	0	0	0	+1	-3	-3	0	-2	-3	-2	+1	0	-4	-2	-3	D	-2	-2	+1	+6	-3	0	+2	-1	-1	-3	-4																					
-1	-3	-3	-1	0	-1	-4	-3	-3	C	0	-3	-3	-3	+9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1	Q	-1	+1	0	0	-3	+5	+2	-2	0	-3	-2	+1	0	-3	-1	0	-1	-2	-1	-2	E	-1	0	0	+2	-4	+2	+5	-2	0	-3	-3	+1	-2	-3																						
-1	0	-1	-3	-2	-2	G	0	-2	0	-1	-3	-2	-2	+6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3	H	-2	0	+1	-1	-3	0	0	-2	+8	-3	-3	-1	-2	-1	-2	-1	-2	-2	+2	-3	I	-1	-3	-3	-3	-1	-3	-3	-4	-3	+4	+2	-3	+1	0	-3	-2																							
-1	-3	-1	+3	L	-1	-2	-3	-4	-1	-2	-3	-4	-3	+2	+4	-2	+2	0	-3	-2	-1	-2	-1	+1	K	-1	+2	0	-1	-3	+1	+1	-2	-1	-3	-2	+5	-1	-3	-1	0	-1	-3	-2	-2	M	-1	-1	-2	-3	-1	0	-2	-3	-2	+1	+2	-1	+5	0	-2	-1	-1																								
-1	-1	+1	F	-2	-3	-3	-3	-2	-3	-3	-3	-3	-1	0	0	-3	0	+6	-4	-2	-2	+1	+3	-1	P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	+7	-1	-1	-4	-3	-2	S	+1	-1	+1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	+4	+1	-3	-2																						
-2	T	0	-1	0	-1	-1	-1	-1	-1	-1	-1	-1	-1	-2	-1	+1	+5	-2	-2	0	W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-2	-1	+1	-4	-3	-2	+11	+2	-3	Y	-2	-2	-2	-3	-2	-1	-2	-3	+2	-1	-1	-2	-1	+3	-3	-2	-2	+2																										
+7	-1	V	0	-3	-3	-3	-1	-2	-2	-3	-3	+3	+1	-2	+1	-1	-2	-2	0	-3	-1	+4																																																																	

[0092] The invention thus contemplates the use of random mutagenesis to identify improved CDRs. In the context of the present invention, conservative substitutions may be defined by substitutions within the classes of amino acids reflected in one or more of Tables 2, 3 and 4: TABLE-US-00002 TABLE 2

Residue	Conservative Substitutions
Asp (D)	Asp (D), Glu (E)
Asn (N)	Asn (N), Gln (Q)
Glu (E)	Glu (E), Asp (D)
Gln (Q)	Gln (Q), Asn (N)
His (H)	His (H), Tyr (Y), Trp (W)
Ile (I)	Ile (I), Val (V), Met (M)
Leu (L)	Leu (L), Val (V), Met (M)
Met (M)	Met (M), Ile (I), Val (V)
Pro (P)	Pro (P), Gly (G)
Ser (S)	Ser (S), Thr (T)
Thr (T)	Thr (T), Ser (S)
Tyr (Y)	Tyr (Y), His (H), Trp (W)
Val (V)	Val (V), Ile (I), Met (M)

Alternative Conservative Amino Acid Residue Substitution Classes:

TABLE-US-00003 TABLE 3

Residue	Conservative Substitutions
Asp (D)	Asp (D), Glu (E)
Asn (N)	Asn (N), Gln (Q)
Glu (E)	Glu (E), Asp (D)
Gln (Q)	Gln (Q), Asn (N)
His (H)	His (H), Tyr (Y), Trp (W)
Ile (I)	Ile (I), Val (V), Met (M)
Leu (L)	Leu (L), Val (V), Met (M)
Met (M)	Met (M), Ile (I), Val (V)
Pro (P)	Pro (P), Gly (G)
Ser (S)	Ser (S), Thr (T)
Thr (T)	Thr (T), Ser (S)
Tyr (Y)	Tyr (Y), His (H), Trp (W)
Val (V)	Val (V), Ile (I), Met (M)

Alternative Physical and Functional Classifications of Amino Acid Residues:

TABLE-US-00004 TABLE 4

Residue	Classification
Ala (A)	Aliphatic
Arg (R)	Basic
Asn (N)	Polar, Uncharged
Asp (D)	Acidic
Cys (C)	Thiol
Glu (E)	Acidic
Gln (Q)	Polar, Uncharged
His (H)	Aromatic
Ile (I)	Aliphatic
Leu (L)	Aliphatic
Met (M)	Aliphatic
Phe (F)	Aromatic
Pro (P)	Aliphatic
Ser (S)	Polar, Uncharged
Thr (T)	Polar, Uncharged
Tyr (Y)	Aromatic
Val (V)	Aliphatic

[0093] More conservative substitutions groupings include: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, and asparagine-glutamine.

[0094] In some embodiments, the hydrophilic amino acid is selected from the group consisting of Arg, Asn, Asp, Gln, Glu, His, Tyr, and Lys.

[0095] Additional groups of amino acids may also be formulated using the principles described in, e.g., Creighton (1984) Proteins: Structure and Molecular Properties (2d Ed. 1993), W. H. Freeman and Company.

[0096] Thus, the sequences of the CDR variants comprised in antibodies or antigen-binding fragments thereof may differ from those of the CDRs of the parent antibody by substitutions, such as substitutions of 4, 3, 2 or 1 amino acid residues. According to an embodiment of the invention, amino acid(s) in the CDR regions may be substituted with conservative substitutions, as defined in 3 Tables above.

[0097] “Homology” or “sequence identity” refers to the percentage of residues in a variant polynucleotide or polypeptide sequence that are identical to those of the non-variant sequence, after alignment of the sequences and introduction of gaps. In specific embodiments, polynucleotide and polypeptide variants have at least about 70%, at least about 75%, at least about 80%, at least about 90%, at least about 95%, at least about 98%, or at least about 99% polynucleotide or polypeptide homology.

[0098] Such variant polypeptide sequences have 70% or higher (i.e., 80%, 85%, 90%, 95%, 97%, 98%, 99% or higher) sequence identity to the sequences recited in the invention. In other embodiments, the invention provides polypeptide fragments comprising contiguous stretches of various lengths of the amino acid sequences disclosed herein. For example, where applicable, the peptide sequences provided herein comprise at least about 5, 10, 15, 20, 30, 40, 50, 75, 100, 150 or more, and all intermediate lengths therebetween, contiguous amino acid residues of one or more sequences disclosed herein.

[0099] The term “treatment” or “treating” as used herein means ameliorating, slowing, attenuating, or reversing the progress or severity of a disease or disorder, or ameliorating, slowing, attenuating, or reversing one or more symptoms or side effects of such disease or disorder. In this invention, “treatment” or “treating” further means an approach for obtaining beneficial or desired clinical results, where “beneficial or desired clinical results” include, without limitation, alleviation of a symptom, diminishment of the extent of a disorder or disease, stabilized (i.e., not worsening) disease or disorder state, delay or slowing of the progression a disease or disorder state, amelioration or palliation of a disease or disorder state, and remission of a disease or disorder, whether partial or total, detectable or undetectable.

[0100] The term “prevention” or “preventing” refers to the administration of antibodies and functional fragments thereof of the invention, thereby preventing or retarding the development of at least one symptom of a disease or condition. The term also includes treating a subject in remission to prevent or hinder relapse.

[0101] Antibodies of the invention may be monoclonal antibodies produced by recombinant DNA techniques.

[0102] The antibody of the invention may be of any isotype. The choice of isotype typically will be guided by the desired effector functions, such as ADCC induction. Exemplary isotypes are IgG1, IgG2, IgG3, and IgG4. Either of the human light chain constant domains, κ or λ , may be used. If desired, the class of an anti-CD47 antibody of the present invention may be switched by known methods. For example, an antibody of the present invention that was originally IgG may be class switched to an IgM antibody of the present invention. Further, class switching techniques may be used to convert one IgG subclass to another, for instance from IgG1 to IgG2. Thus, the effector function of the antibodies of the present invention may be changed by isotype switching to, e.g., an IgG1, IgG2, IgG3, IgG4, IgD, IgA, IgE, or IgM antibody for various therapeutic uses. In one embodiment an antibody of the present invention is an IgG4 antibody. An antibody is said to be of a particular isotype if its amino acid sequence is most homologous to that isotype, relative to other isotypes.

[0103] In some embodiments, the antibodies of the invention are full-length antibodies, preferably IgG antibodies. In other embodiments, the antibodies of the invention are antibody antigen-binding fragments or single chain antibodies.

[0104] In some embodiments, the anti-CD47 antibody is a monovalent antibody, preferably a monovalent antibody as described in WO2007059782 (which is incorporated herein by reference in its entirety) having a deletion of the hinge region. Accordingly, in some embodiments, the antibody is a monovalent antibody, wherein said anti-CD47 antibody is constructed by a method comprising: i) providing a nucleic acid construct encoding the light chain of said monovalent antibody, said construct comprising a nucleotide sequence encoding the VL region of a selected antigen specific anti-CD47 antibody and a nucleotide sequence encoding the constant CL region of an Ig, wherein said nucleotide sequence encoding the VL region of a selected antigen specific antibody and said nucleotide sequence encoding the CL region of an Ig are operably linked together, and wherein, in case of an IgG1 subtype, the nucleotide sequence encoding the CL region has been modified such that the CL region does not contain any amino acids capable of forming disulfide bonds or covalent bonds with other peptides comprising an identical amino acid sequence of the CL region in the presence of polyclonal human IgG or when administered to an animal or human being; ii) providing a nucleic acid construct encoding the heavy chain of said monovalent antibody, said construct comprising a nucleotide sequence encoding the VH region of a selected antigen specific antibody and a nucleotide sequence encoding a constant CH region of a human Ig, wherein the nucleotide sequence encoding the CH region has been modified such that the region corresponding to the hinge region and, as required by the Ig subtype, other regions of the CH region, such as the CH3 region, does not comprise any amino acid residues which participate in the formation of disulphide bonds or covalent or stable non-covalent inter-heavy chain bonds with other peptides comprising an identical amino acid sequence of the CH region of the human Ig in the presence of polyclonal human IgG or when administered to an animal human being, wherein said nucleotide sequence encoding the VH region of a selected antigen specific antibody and said nucleotide sequence encoding the CH region of said Ig are operably linked together; iii) providing a cell expression system for producing said monovalent antibody; iv) producing said monovalent antibody by co-expressing the nucleic acid constructs of (i) and (ii) in cells of the cell expression system of (iii).

[0105] Similarly, in some embodiments, the anti-CD47 antibody of the invention is a monovalent antibody, which comprises: [0106] (i) a variable domain of an antibody of the invention as

described herein or an epitope-binding part of the said domain, and [0107] (ii) a CH domain of an immunoglobulin or a domain thereof comprising the CH2 and CH3 domains, wherein the CH domain or domain thereof has been modified such that the domain corresponding to the hinge domain and, if the immunoglobulin is not an IgG4 subtype, other domains of the CH domain, such as the CH3 domain, do not comprise any amino acid residues, which are capable of forming disulfide bonds with an identical CH domain or other covalent or stable non-covalent inter-heavy chain bonds with an identical CH domain in the presence of polyclonal human IgG.

[0108] In some other embodiments, the heavy chain of the monovalent antibody of the invention has been modified such that the entire hinge region has been deleted.

[0109] In other embodiments, the sequence of the monovalent antibody has been modified so that it does not comprise any acceptor sites for N-linked glycosylation.

[0110] The invention also includes "Bispecific Antibodies," wherein an anti-CD47 binding region (e.g., a CD47-binding region of an anti-CD47 monoclonal antibody) is part of a bivalent or polyvalent bispecific scaffold that targets more than one epitope, (for example a second epitope could comprise an epitope of an active transport receptor, such that the bispecific antibody would exhibit improved transcytosis across a biological barrier, such as the Blood Brain Barrier). Thus, in other embodiments, the monovalent Fab of an anti-CD47 antibody may be joined to an additional Fab or scfv that targets a different protein to generate a bispecific antibody. A bispecific antibody can have a dual function, for example a therapeutic function imparted by an anti-CD47 binding domain and a transport function that can bind to a receptor molecule to enhance transfer cross a biological barrier, such as the blood brain barrier.

[0111] Antibodies and epitope-binding fragments thereof of the invention also include single chain antibodies. Single chain antibodies are peptides in which the heavy and light chain Fv domains are connected. In some embodiment, the present invention provides a single-chain Fv (scFv) wherein the heavy and light chains in the Fv of an anti-CD47 antibody of the present invention are joined with a flexible peptide linker (typically of about 10, 12, 15 or more amino acid residues) in a single peptide chain. Methods of producing such antibodies are described in for instance U.S. Pat. No. 4,946,778, Pluckthun in *The Pharmacology of Monoclonal Antibodies*, vol. 113, Rosenberg and Moore eds. Springer-Verlag, New York, pp. 269-315 (1994), Bird et al., *Science* 242, 423-426 (1988), Huston et al., *PNAS USA* 85, 5879-5883 (1988) and McCafferty et al., *Nature* 348, 552-554 (1990). The single chain antibody may be monovalent, if only a single VH and VL are used, bivalent, if two VH and VL are used, or polyvalent, if more than two VH and VL are used.

[0112] Antibodies of the invention are produced by any technique known in the art, such as, but not limited to, any chemical, biological, genetic, or enzymatic technique, alone or in combination. In general, if knowing the amino acid sequence of a desired sequence, one skilled in the art can readily generate such an antibody by standard techniques used to generate polypeptides. For example, these antibodies can be synthesized using well-known solid-phase methods, preferably using a commercially available peptide synthesis apparatus (such as that manufactured by Applied Biosystems, Foster City, California) and following the manufacturer's instructions. Alternatively, antibodies of the invention can be synthesized by recombinant DNA techniques well known in the art. For example, after incorporating a DNA sequence encoding an antibody into an expression vector and introducing these vectors into eukaryotic or prokaryotic host cells suitable for expressing the desired antibodies, antibodies as DNA expression products can be obtained, which can then be isolated from the host cells using known techniques.

[0113] The antibodies and epitope-binding fragments thereof described herein may be modified by inclusion of any suitable number of modified amino acids and/or associations with such conjugated substituents. Suitability in this context is generally determined by the ability to retain the CD47 selectivity and/or the CD47 specificity at least substantially associated with the non-derivatized parent anti-CD47 antibody. The inclusion of one or more modified amino acids may be advantageous in, for example, increasing polypeptide serum half-life, reducing polypeptide

antigenicity, or increasing polypeptide storage stability. Amino acid(s) are modified, for example, co-translationally or post-translationally during recombinant production (e.g., N-linked glycosylation at N-X-S/T motifs during expression in mammalian cells) or modified by synthetic means. Non-limiting examples of a modified amino acid include a glycosylated amino acid, a sulfated amino acid, a prenylated (e.g., farnesylated, geranyl-geranylated) amino acid, an acetylated amino acid, an acylated amino acid, a PEGylated amino acid, a biotinylated amino acid, a carboxylated amino acid, a phosphorylated amino acid, and the like. References adequate to guide one of skill in the modification of amino acids are replete throughout the literature, see e.g., Walker (1998) Protein Protocols On CD-Rom, Humana Press, Totowa, NJ. The modified amino acid may, for instance, be selected from a glycosylated amino acid, a PEGylated amino acid, a farnesylated amino acid, an acetylated amino acid, a biotinylated amino acid, an amino acid conjugated to a lipid moiety, or an amino acid conjugated to an organic derivatizing agent.

[0114] The antibodies and antigen-binding fragments thereof of the invention may also be chemically modified by covalent conjugation to a polymer to for instance increase their circulating half-life. Exemplary polymers and methods to attach them to peptides are illustrated in for instance U.S. Pat. Nos. 4,766,106; 4,179,337; 4,495,285 and 4,609,546. Additional illustrative polymers include polyoxyethylated polyols and polyethylene glycol (PEG) (e.g., a PEG with a molecular weight of between about 1,000 and about 40,000 D, such as between about 2,000 and about 20,000 D, e.g., about 3,000-12,000 D).

[0115] The term “subject” refers to a warm-blooded animal, preferably a mammal (including humans, domestic and farm animals, zoo animals, sport animals, or pet animals such as dogs, cats, cows, horses, sheep, pigs, goats, rabbits, etc.), more preferably humans. In one embodiment, a subject may be a “patient”, i.e., a warm-blooded animal, more preferably a human, who is waiting to receive or is receiving medical care or will be a subject of a medical procedure, or monitoring for the development of a disease. In one embodiment, the subject is an adult (e.g., a subject over 18 years old). In another embodiment, the subject is a child (e.g., a subject under 18 years old). In one embodiment, the subject is a male. In another embodiment, the subject is a female.

[0116] In one embodiment of the invention, a sample is a biological sample. Examples of biological samples include, but are not limited to, tissue lysates and extracts prepared from diseased tissues, body fluids, preferably blood, more preferably serum, plasma, synovial fluid, bronchoalveolar lavage fluid, sputum, lymph fluid, ascites, urine, amniotic fluid, peritoneal fluid, cerebrospinal fluid, pleural fluid, pericardial fluid, and alveolar macrophages.

[0117] In one embodiment of the invention, the term “sample” is intended to mean a sample taken from an individual prior to any analysis.

[0118] Accordingly, in some embodiments, the anti-CD47 antibodies and antigen-binding fragments thereof of the invention include intact antibodies, such as IgG (IgG1, IgG2, IgG3 and IgG4 subtypes), IgA (IgA1 and IgA2 subtypes), IgM and IgE; antigen-binding fragments such as SDR, CDR, Fv, dAb, Fab, Fab.sub.2, Fab', F(ab').sub.2, Fd, scFv, camelid antibodies or nanobodies; variant sequences of antibodies or antigen-binding fragments thereof, such as variant sequences having at least 80% sequence identity to the above antibodies or antigen-binding fragments thereof. In some embodiments, the invention also includes derivatives comprising the anti-CD47 antibody or antigen-binding fragment thereof, such as chimeric antibodies, humanized antibodies, fully human antibodies, recombinant antibodies, and bispecific antibodies derived from an intact antibody.

[0119] In a further aspect, the invention relates to an expression vector comprising nucleotide sequences encoding one or more polypeptide chains of the antibody or antigen-binding fragment thereof of the invention. Such expression vectors can be used to recombinantly produce the antibodies and antigen-binding fragments thereof of the invention.

[0120] An expression vector in the context of the present invention may be any suitable DNA or RNA vector, including chromosomal, non-chromosomal, and synthetic nucleic acid vectors (a

nucleic acid sequence comprising a suitable set of expression control elements). Examples of such vectors include derivatives of SV40, bacterial plasmids, phage DNA, baculovirus, yeast plasmids, vectors derived from combinations of plasmids and phage DNA, and viral nucleic acid (RNA or DNA) vectors. In some embodiments, an anti-CD47 antibody-encoding nucleic acid is comprised in a naked DNA or RNA vector, including, for example, a linear expression element (as described in, for instance, Sykes and Johnston, *Nat Biotech* 12, 355-59 (1997)), a compacted nucleic acid vector (as described in for instance U.S. Pat. No. 6,077,835 and/or WO 00/70087), a plasmid vector such as pBR322, pUC 19/18, or pUC 118/119, a minimally-sized nucleic acid vector (as described in, for instance, Schakowski et al., *Mol Ther* 3, 793-800 (2001)), or as a precipitated nucleic acid vector construct, such as a CaPO₄-precipitated construct (as described in, for instance, WO 00/46147, Benvenisty and Reshef, *PNAS USA* 83, 9551-55 (1986), Wigler et al., *Cell* 14, 725 (1978), and Coraro and Pearson, *Somatic Cell Genetics* 2, 603 (1981)). Such nucleic acid vectors and the usage thereof are well known in the art (see for instance U.S. Pat. Nos. 5,589,466 and 5,973,972).

[0121] In some embodiments, the vector is suitable for expression of anti-CD47 antibodies or epitope-binding fragments thereof of the invention in a bacterial cell. Examples of such vectors include expression vectors such as BlueScript (Stratagene), pIN vectors (Van Heeke & Schuster, *J. Biol. Chem.* 264, 5503-5509 (1989), pET vectors (Novagen, Madison, Wisconsin), and the like.

[0122] An expression vector may also be a vector suitable for expression in a yeast system. Any vector suitable for expression in a yeast system may be employed. Suitable vectors include, for example, vectors comprising constitutive or inducible promoters such as alpha factor, alcohol oxidase and PGH (reviewed in: F. Ausubel et al., ed. *Current Protocols in Molecular Biology*, Greene Publishing and Wiley InterScience New York (1987), Grant et al., *Methods in Enzymol* 153, 516-544 (1987), Mattanovich, D. et al. *Methods Mol. Biol.* 824, 329-358 (2012), Celik, E. et al. *Biotechnol. Adv.* 30(5), 1108-1118 (2012), Li, P. et al. *Appl. Biochem. Biotechnol.* 142(2), 105-124 (2007), Böer, E. et al. *Appl. Microbiol. Biotechnol.* 77(3), 513-523 (2007), van der Vaart, J. M. *Methods Mol. Biol.* 178, 359-366 (2002), and Holliger, P. *Methods Mol. Biol.* 178, 349-357 (2002)).

[0123] In an expression vector of the invention, the anti-CD47 antibody-encoding nucleic acids may comprise or be associated with any suitable promoter, enhancer, and other expression-facilitating elements. Examples of such elements include strong expression promoters (e.g., human CMV IE promoter/enhancer as well as RSV, SV40, SL3-3, MMTV, and HIV LTR promoters), effective poly (A) termination sequences, an origin of replication for plasmid product in *E. coli*, an antibiotic resistance gene as selectable marker, and/or a convenient cloning site (e.g., a polylinker). Nucleic acids may also comprise an inducible promoter as opposed to a constitutive promoter such as CMV IE (the skilled artisan will recognize that such terms are actual descriptors of a degree of gene expression under certain conditions).

[0124] In an even further aspect, the invention relates to a recombinant eukaryotic or prokaryotic host cell, such as a transfectoma, which produces an antibody or epitope-binding fragment thereof of the invention as defined herein or a bispecific molecule of the invention as defined herein. Examples of host cells include yeast, bacteria, and mammalian cells, such as CHO or HEK cells. For example, in some embodiments, the present invention provides a cell comprising a nucleic acid stably integrated into the cellular genome that comprises a sequence coding for expression of an anti-CD47 antibody of the present invention or an epitope-binding fragment thereof. In other embodiments, the present invention provides a cell comprising a non-integrated nucleic acid, such as a plasmid, cosmid, phagemid, or linear expression element, which comprises a sequence coding for expression of an anti-CD47 antibody or epitope-binding fragment thereof of the invention.

[0125] Antibodies and antigen-binding fragments thereof of the invention can be produced in different cell lines, such as human cell lines, non-human mammalian cell lines, and insect cell lines, such as CHO cell lines, HEK cell lines, BHK-21 cell lines, murine cell lines (such as

myeloma cell line), fibrosarcoma cell line, PER.C6 cell line, HKB-11 cell line, CAP cell line, and HuH-7 human cell line (Dumont et al., 2015, Crit Rev Biotechnol., September 18, 1-13, the contents of which are incorporated herein by reference).

[0126] The antibodies of the invention are suitably separated from the culture medium by conventional immunoglobulin purification methods, such as Protein A-Sepharose, hydroxyapatite chromatography, gel electrophoresis, dialysis, or affinity chromatography.

[0127] The present invention further relates to a composition comprising, consisting of, or consisting essentially of the antibodies of the invention.

[0128] As used herein, the term “consisting essentially of” with respect to a composition means that at least one antibody of the invention as described above is the only biologically active therapeutic agent or reagent in the composition.

[0129] In one embodiment, the composition of the invention is a pharmaceutical composition and further comprises a pharmaceutically acceptable excipient, diluent, or carrier.

[0130] The term “pharmaceutically acceptable carrier” refers to an excipient that does not produce adverse, allergic, or other adverse reactions when administered to animals, preferably humans, including any, and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents and the like. For human administration, preparations should meet sterility, pyrogenicity, general safety and purity standards as required by regulatory agencies (e.g., FDA or EMA).

[0131] The invention further relates to a medicament comprising, consisting of, or consisting essentially of the antibodies of the invention.

[0132] In some embodiments, the glycosylation of the antibodies of the invention is modified. For example, an aglycosylated antibody can be made (i.e., the antibody lacks glycosylation).

Glycosylation can be altered to, for example, increase the affinity of the antibody for an antigen or change the ADCC activity of the antibody. Such carbohydrate modifications can be accomplished by, for example, altering one or more sites of glycosylation within the antibody sequence. For example, one or more amino acid substitutions can be made that result in elimination of one or more variable region framework glycosylation sites to thereby eliminate glycosylation at that site. Such aglycosylation may increase the affinity of the antibody for an antigen. Such an approach is described in further detail in U.S. Pat. Nos. 5,714,350 and 6,350,861 by Co et al. Additionally, or alternatively, an antibody can be made that has an altered type of glycosylation, such as a hypofucosylated or an afucosylated antibody having reduced amounts of fucosyl residues or having no fucosyl residues, or an antibody having increased bisecting GlcNAc structures. Such altered glycosylation patterns have been demonstrated to increase the ADCC ability of antibodies. Such carbohydrate modifications can be accomplished by, for example, expressing the antibody in a host cell with altered glycosylation machinery. Cells with altered glycosylation machinery have been described in the art and can be used as host cells in which to express recombinant antibodies of the invention to thereby produce an antibody with altered glycosylation. For example, EP 1,176,195 by Hang et al. describes a cell line with a functionally disrupted FUT8 gene, which encodes a fucosyl transferase, such that antibodies expressed in such a cell line exhibit hypofucosylation. PCT Publication WO 03/035835 by Presta describes a variant CHO cell line, Lec13 cells, with reduced ability to attach fucose to Asn (297)-linked carbohydrates, also resulting in hypofucosylation of antibodies expressed in that host cell (see also Shields, R. L. et al., (2002), J. Biol Chem. 277: 26733-26740). PCT Publication WO 99/54342 by Umana et al., describes cell lines engineered to express glycoprotein-modifying glycosyl transferases (e.g., beta(1,4)-N-acetylglucosaminyltransferase III (GnTIII)) such that antibodies expressed in the engineered cell lines exhibit increased bisecting GlcNAc structures which results in increased ADCC activity of the antibodies (see also Umana et al., (1999), Nat. Biotech. 17:176-180). Eureka Therapeutics further describes genetically engineered CHO mammalian cells capable of producing antibodies with altered mammalian glycosylation patterns lacking fucosyl residues

(<http://www.eureka-inc.com/a&boutus/companyoverview.html>). Alternatively, the human antibodies (preferably monoclonal antibodies) of the invention can be produced in yeast or filamentous fungi which are used in the production of antibodies having mammalian-like glycosylation patterns and capable of producing a glycosylation model lacking fucose (see e.g., EP1297172B1).

[0133] The antibody of the invention acts on human CD47, can specifically recognize the expression of CD47 on the surface of human tumor cells, thereby blocking the inhibitory signal pathway transmitted by the binding of CD47 to the receptor SIRP α , and can promote the phagocytosis of tumor cells mediated by macrophages, thus providing a novel method for tumor immunotherapy. The antibody of the invention can be used to prevent and/or treat autoimmune diseases, such as arthritis, rheumatoid arthritis, psoriasis, multiple sclerosis, ulcerative colitis, Crohn's disease, systemic lupus erythematosus, glomerulonephritis, dilated cardiomyopathy-like disease, Sjogren's syndrome, allergic contact dermatitis, polymyositis, scleroderma, periarteritis polyarteritis, rheumatic fever, vitiligo, insulin-dependent diabetes mellitus, Behcet's syndrome and chronic thyroiditis, etc., immune response to grafts, allergies, infectious diseases, neurodegenerative diseases such as Parkinson's disease, Huntington's disease, Machado-Joseph disease, muscular atrophy lateral sclerosis and Creutzfeldt-Jakob disease, etc., and tumors, such as leukemia, lymphoma, myeloma, brain tumor, squamous cell carcinoma of the head and neck, non-small cell lung cancer, nasopharyngeal cancer, esophageal cancer, gastric cancer, pancreatic cancer, gallbladder cancer, liver cancer, colorectal cancer, breast cancer, ovarian cancer, cervical cancer, endometrial cancer, uterine sarcoma, prostate cancer, bladder cancer, renal cell carcinoma and melanoma, etc.

[0134] In some embodiments, the invention relates to the following technical solutions.

[0135] Item 1. An isolated anti-human CD47 antibody, an antigen-binding fragment, or a variant thereof, wherein said antibody or antigen-binding fragment thereof comprises a light chain variable region and/or a heavy chain variable region, wherein [0136] the amino acid sequences of LCDR1, LCDR2, and LCDR3 of the light chain variable region are set forth in SEQ ID Nos. 1, 2, and 3, respectively; and/or [0137] the amino acid sequence of HCDR1 of the heavy chain variable region is set forth in SEQ ID No. 4, the amino acid sequence of HCDR2 of the heavy chain variable region has at least about 94% identity to SEQ ID No. 5 or 6; the amino acid sequence of HCDR3 of the heavy chain variable region has at least about 78% identity to SEQ ID No. 7 or 8; or [0138] the amino acid sequences of the CDRs of the light chain variable region or the heavy chain variable region have at least 70% identity to the sequences set forth in SEQ ID Nos. 1-8 respectively and retain the biological activity of the corresponding parent sequence; or [0139] the amino acid sequences of the CDRs of the light chain variable region or the heavy chain variable region are variant sequences of the sequences set forth in SEQ ID Nos. 1-8 respectively obtained after deletion, substitution and/or addition of one or more amino acid residues which retain the biological activities of the corresponding parent sequences, [0140] wherein the variant is one selected from the group consisting of chimeric antibodies, humanized antibodies, and fully human antibodies.

[0141] Item 2. The anti-human CD47 antibody, antigen-binding fragment or variant thereof according to item 1, wherein the heavy chain constant region sequence of the antibody is the constant region sequence of one selected from the group consisting of human IgG1, IgG2, IgG3, IgG4, IgA, IgM, IgE, IgD, and/or, the light chain constant region sequence of the antibody is the constant region sequence of κ chain or λ chain; [0142] preferably, the heavy chain constant region sequence is the constant region sequence of IgG1 or IgG4, and/or the light chain constant region sequence is the constant region sequence of κ light chain.

[0143] Item 3. The anti-human CD47 antibody, antigen-binding fragment or variant thereof according to item 1 or 2, wherein the amino acid sequence of the light chain variable region of a CD47 chimeric antibody or a functional fragment thereof is as set forth in SEQ ID NO. 9 or 10; or has at least 70% identity to the sequence set forth in SEQ ID No. 9 or 10 and retains the biological

activity of the corresponding parent sequence; or is a variant sequence of the sequence set forth in SEQ ID NO. 9 or 10 obtained by deleting, substituting and/or adding one or more amino acid residues which retains the biological activity of the corresponding parent sequence; and/or the amino acid sequence of the heavy chain variable region is as set forth in SEQ ID NO. 11 or 12; or has at least 70% identity to the sequence set forth in SEQ ID NO. 11 or 12 and retains the biological activity of corresponding parent sequence; or is a variant sequence of the sequence set forth in SEQ ID NO. 11 or 12 obtained by deleting, substituting and/or adding one or more amino acid residues that retains the biological activity of the corresponding parent sequence; and/or [0144] the amino acid sequences of the light chain constant region and the heavy chain constant region of the chimeric CD47 antibody or a functional fragment thereof are set forth in SEQ ID NO. 13 and SEQ ID Nos. 14-16 respectively; or have at least 70% identity to the sequences set forth in SEQ ID NO. 13 and SEQ ID Nos. 14-16 respectively and retain the biological activity of the corresponding parent sequences; or are variant sequences of the sequences set forth in SEQ ID NO. 13 and SEQ ID Nos. 14-16 respectively obtained by deleting, substituting and/or adding one or more amino acid residues that retain the biological activity of the corresponding parent sequences. [0145] Item 4. The anti-human CD47 antibody, antigen-binding fragment or variant thereof according to item 1 or 2, wherein the framework regions of the light chain variable region of the anti-human CD47 antibody, antigen-binding fragment or variant thereof include FR-L1, FR-L2, FR-L3 and FR-L4, and the framework regions of the heavy chain variable include FR-H1, FR-H2, FR-H3 and FR-H4, and/or wherein, [0146] the amino acid sequence of the FR-L1 is set forth in SEQ ID NO. 17; [0147] the amino acid sequence of the FR-L2 is set forth in SEQ ID NO. 18, or an amino acid sequence as set forth in SEQ ID NO. 18 obtained by one of the following substitutions or any combination thereof: [0148] the 12th amino acid R is substituted with T; [0149] the amino acid sequence of the FR-L3 is set forth in SEQ ID NO. 19, or an amino acid sequence as set forth in SEQ ID NO. 19 obtained by one of the following substitutions or any combination thereof: [0150] the 31st amino acid Y is substituted with F; [0151] the amino acid sequence of the FR-L4 is set forth in SEQ ID NO. 20; [0152] the amino acid sequence of the FR-H1 is set forth in SEQ ID NO. 21; [0153] the amino acid sequence of the FR-H2 is set forth in SEQ ID NO. 22; [0154] the amino acid sequence of the FR-H3 is set forth in SEQ ID NO. 23, or an amino acid sequence as set forth in SEQ ID NO. 23 obtained by one of the following substitutions or any combination thereof: [0155] the 8th amino acid E is substituted with T, [0156] the 11th amino acid S is substituted with N, [0157] the 31st amino acid A is substituted with V; and/or [0158] the amino acid sequence of the FR-H4 is set forth in SEQ ID NO. 24.

[0159] Item 5. The anti-human CD47 antibody, antigen-binding fragment or variant thereof according to item 1 or 2, wherein the amino acid sequence of the light chain variable region is as set forth in any one of SEQ ID Nos. 25-27, and/or the amino acid sequence of the heavy chain variable region is as set forth in any one of SEQ ID Nos. 28-33; or is a variant sequence having at least 70% identity to one of the amino acid sequences set forth SEQ ID Nos. 25-27 or 28-33 and retaining the biological activity of the corresponding parent sequence; or is a variant sequence of one of the sequences set forth in SEQ ID Nos. 25-27 or 28-33 obtained after deletion, substitution and/or addition of one or more amino acid residues which retains the biological activity of the corresponding parent sequence; [0160] preferably, the light chain variable region sequence is as set forth in SEQ ID NO. 26 or 27, and/or the heavy chain variable region sequence is as set forth in SEQ ID NO. 33.

[0161] Item 6. The anti-human CD47 antibody, antigen-binding fragment or variant thereof according to any one of items 1-5, wherein the antigen-binding fragment is one or more selected from the group consisting of F(ab').sub.2, Fab', Fab, Fd, Fv, scFv, diabody, camelid antibody, CDR, and antibody minimal recognition unit (dAb), preferably, the antigen-binding fragment is Fab, F(ab').sub.2 or scFv.

[0162] Item 7. An isolated nucleic acid molecule selected from the group consisting of: [0163] (1)

DNA or RNA encoding the anti-human CD47 antibody, antigen-binding fragment or variant thereof according to any one of items 1-6; [0164] (2) a nucleic acid that is completely complementary to the nucleic acid defined in (1).

[0165] Item 8. A vector comprising the nucleic acid molecule according to item 7 operably linked, preferably, the vector is an expression vector.

[0166] Item 9. A host cell comprising the nucleic acid molecule according to item 7 or the vector according to item 8.

[0167] Item 10. A composition comprising the anti-human CD47 antibody, antigen-binding fragment or variant thereof according to any one of items 1-6, the nucleic acid molecule according to item 7, the vector according to item 8, or the host cell according to item 9, and a pharmaceutically acceptable excipient, diluent and/or carrier.

[0168] Item 11. A method of producing the anti-human CD47 antibody, antigen-binding fragment or variant thereof according to any one of items 1-6, comprising the steps of: [0169] culturing the host cells according to item 9 under culture conditions suitable for the expression of the anti-human CD47 antibody, antigen-binding fragment or variant thereof, and optionally, isolating and purifying the obtained products.

[0170] Item 12. Use of the anti-human CD47 antibody, antigen-binding fragment or variant thereof according to any one of items 1-6, the nucleic acid molecule according to item 7, the vector according to item 8, or the host cell according to item 9 in the manufacture of a medicament for the prevention and/or the treatment of autoimmune diseases, immune responses against grafts, allergies, infectious diseases, neurodegenerative diseases and tumors; [0171] preferably, the autoimmune diseases are one or more selected from the group consisting of arthritis, rheumatoid arthritis, psoriasis, multiple sclerosis, ulcerative colitis, Crohn's disease, systemic lupus erythematosus, glomerulonephritis, dilated cardiomyopathy-like disease, Sjogren's syndrome, allergic contact dermatitis, polymyositis, scleroderma, periarteritis polyarteritis, rheumatic fever, vitiligo, insulin-dependent diabetes mellitus, Behcet's syndrome and chronic thyroiditis; [0172] preferably, the immune response against the graft is graft versus host disease; preferably, the allergies are one of more selected from the group consisting of urticaria, eczema, angioedema, allergic rhinitis, bronchial asthma, laryngeal edema, food allergic gastroenteritis, and anaphylactic shock; [0173] preferably, the infectious disease refers to the local and systemic inflammatory response caused by the invasion of viruses, bacteria, fungi, parasites and other pathogens and specific toxins into the human body; [0174] preferably, the neurodegenerative diseases are one or more selected from the group consisting of Parkinson's disease, Huntington's disease, Machado-Joseph disease, amyotrophic lateral sclerosis, Creutzfeldt-Jakob disease; [0175] preferably, the tumors are one or more selected from the group consisting of leukemia, lymphoma, myeloma, brain tumor, head and neck squamous cell carcinoma, non-small cell lung cancer, nasopharyngeal cancer, esophageal cancer, gastric cancer, pancreatic cancer, gallbladder cancer, liver cancer, colorectal cancer, breast cancer, ovarian cancer, cervical cancer, endometrial cancer, uterine sarcoma, prostate cancer, bladder cancer, renal cell carcinoma, and melanoma.

[0176] Where a range of values is provided, it is understood that unless the context clearly dictates otherwise, each intervening value, to the tenth of the unit of the lower limit, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both the limits, ranges excluding either or both of those included limits are also included in the invention.

[0177] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials similar or equivalent to those described herein can

also be used in the practice or testing of the present invention, the preferred methods and materials are now described. All publications mentioned herein are incorporated herein by reference.

[0178] On the one hand, the examples provided by the invention are used to describe the preparation processes of the antibodies of the invention. The preparation processes are only to illustrate the relevant methods and is not limiting. It is well known to those skilled in the art that the invention can be modified without departing from the spirit of the invention, and such modifications also fall within the scope of the invention. On the other hand, the invention provides examples to illustrate the characteristics and advantages of the antibodies of the invention, but the invention is not limited to these characteristics and advantages.

[0179] Embodiments of the invention will be illustrated in detail by following examples, but those skilled in the art will understand that the following examples are only for illustrating the invention, and should not be construed as limiting the scope of the invention. Those not indicating the specific conditions in the examples are carried out according to the conventional conditions or the conditions suggested by the manufacturer. The reagents or instruments used of which the manufacturers are not specified are all conventional products that could be commercially purchased.

EXAMPLES

Example 1. Preparation of the Murine Anti-Human CD47 Monoclonal Antibodies

1.1. Animal Immunizations

[0180] The 6-8-week-old female wild-type Balb/c mice (Beijing Vital River) were selected as the experimental material. After the mice acclimated to the environment for one week, the immunization began. For the first immunization, 50 μ g of the recombinant human CD47-Fc protein (ACROBiosystems, catalog number CD7-H52A5) and complete Freund's adjuvant (Sigma-Aldrich, catalog number F5881) were thoroughly mixed to form an emulsion and injected intraperitoneally into the mice. Two weeks later, a booster immunization was given. For the booster immunization, 25 μ g of the recombinant human CD47-Fc protein and incomplete Freund's adjuvant (Sigma-Aldrich, catalog number F5806) were thoroughly mixed to form an emulsion and injected intraperitoneally into the mice. The immunization was boosted with the same process every 2 weeks, a total of 3 times. On day 10 after the last immunization, blood was collected from the posterior orbital venous plexus of the mice, centrifuged to isolate serum, and the antibody titer against the recombinant human CD47-Fc protein was determined with ELISA. Mice with high titers were selected for fusion to make hybridomas. Three days before fusion, 50 μ g of the recombinant human CD47-Fc protein was injected intraperitoneally without an adjuvant. On the day of fusion, the spleen was taken aseptically, prepared into individual splenocytes suspension with DMEM medium (Dulbecco's Modified Eagle Medium, Gibco, catalog number 11965) comprising 15% fetal bovine serum (FBS) (Corning, catalog number 35-076-CV), 0.25 μ M CpG ODN (InvivoGen, catalog number tlrl-1826-1) was added to stimulate for 24 hours, BTX ECM2001 electrofusion instrument (Harvard Apparatus, catalog number 45-0010) was used for electrofusion.

1.2. Preparation of Hybridomas

[0181] The myeloma cells SP2/0 (purchased from the Cell Bank of Chinese Academy of Sciences) grew to logarithmic phase, centrifuged at 1000 rpm for 5 min, the supernatant was discarded, the cells were suspended with incomplete DMEM culture medium and counted, the required number of cells were taken, washed twice with E-fusion buffer composed of 0.3 M mannitol (Sigma, catalog number M8429), 0.1 mM ammonium sulfate (Sigma, catalog number M7774) and 0.1 mM calcium chloride ((Sigma, catalog number C5670), centrifuged at 1000 rpm for 5 min. At the same time, the splenocytes were washed twice with E-fusion buffer, centrifuged at 240 g for 6 min. The myeloma cells and splenocytes were mixed together according to a certain ratio, washed once with E-fusion buffer in a 50 ml plastic centrifuge tube, centrifuged at 240 g for 6 min. The supernatant was discarded and tapping the bottom of the centrifuge tubes with your hand to loosen the precipitated

cells evenly; E-fusion buffer was added according to a ratio of 2 ml E-fusion buffer/(1-1.5)*10E7 splenocytes, 2 ml of cell mixture was added into each fusion well with a gap of 10 mm (Harvard Apparatus, catalog number 45-0107), the parameter settings were checked, and Automatic Start was pressed. After fusion, the cell suspension was carefully transferred from the fusion well to a 50 ml centrifuge tube containing 28 ml of HAT medium (DMEM+HAT, Sigma, catalog number: H0262-10VL) with a pipette under sterile conditions, let stand at 37° C. for 10 min, HAT medium was added according to a ratio of (3-5)*10E4 cells per well and plated, and the cells were cultured at 37° C. in a 5% CO₂ incubator. After 5 days, ½ medium was replaced with HAT medium. After 7~10 days, HAT medium was replaced with HT medium (DMEM+HT, Sigma, catalog number H0137-10VL). The growth of the hybridomas was regularly observed, and the supernatant was pipetted when the hybridomas grew to more than 1/10 of the area of the bottom of the well for antibody detection. The cells of the positive clones were expanded and frozen.

1.3. Screening and Identification of Clones

[0182] ELISA was used to screen the anti-human CD47 antibodies in the hybridoma culture supernatants. A 96-well high-adsorption ELISA plate (Costar, catalog number 42592) was coated with 1 µg/mL of the recombinant human CD47 (Sino Biological Inc.) at 100 µL per well using carbonate buffer solution, pH=9.6 at 4° C. overnight. The plate was washed 5 times with PBST, and then blocked using PBST with 1% BSA, 300 µL/well, and incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The culture supernatant samples and positive serum control were added, 100 µL/well, incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The horseradish peroxidase-labeled anti-mouse IgG antibody (Abcam, catalog number Ab7068) diluted 1:10000 in PBST with 1% BSA was then added, 100 µL per well, and incubated at 25° C. for 1 hour. The plate was washed 5 times with PBST. The colorimetric substrate TMB was added, 100 µL/well, and developed at room temperature for 10 minutes. 1M H₂SO₄ was added, 100 µL/well, to stop the development. The absorbance at 450 nm was read on a microplate reader. Positive clones capable of secreting the anti-human CD47 selection antibodies were selected according to the magnitude of OD_{450nm}.

[0183] ELISA was used to determine whether the anti-human CD47 selection antibodies secreted by the positive clones can block the binding of CD47/SIRPα. A 96-well high-adsorption ELISA plate was coated with 1 µg/mL of the recombinant human CD47-Fc (ACROBiosystems) at 100 µL per well using carbonate buffer solution, pH=9.6 at 4° C. overnight. The plate was washed 5 times with PBST, and then blocked using PBST with 1% BSA, 300 µL/well, and incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The anti-human CD47 selection antibody samples were added, 50 µL/well, the biotin-labelled SIRPα (ACROBiosystems) was added at a concentration of 4 nM (a final concentration of 2 nM), 50 µL/well, incubated at 25° C. for 90 minutes, and then washed 5 times with PBST. The Streptavidin-HRP (BD Pharmingen, catalog number 554066) diluted 1:1000 in PBST with 1% BSA was then added, 100 µL per well, and incubated at 25° C. for 1 hour. The plate was washed 5 times with PBST. The colorimetric substrate TMB was added, 100 µL/well, and developed at room temperature for 10 minutes. 1M H₂SO₄ was added, 100 µL/well, to stop the development. The absorbance at 450 nm was read on a microplate reader. The anti-human CD47 selection antibodies capable of inhibiting the binding of the human CD47-Fc/biotin-labelled SIRPα have a neutralizing activity. The positive clones capable of secreting the anti-human CD47 neutralizing antibodies were selected according to the magnitude of the blocking capability.

[0184] Red blood cells (RBC) highly express CD47, and some anti-CD47 antibodies can cause erythrocytes agglutination. Whether the anti-human CD47 neutralizing antibodies secreted by the positive clones cause erythrocytes agglutination was determined. Anticoagulated healthy human whole blood donated internally was added in DPBS (Gibco, catalog number 14190250), centrifuged, the supernatant was discarded, the red blood cells at the bottom of the tube were collected, washed 3 times, and the red blood cells were resuspended in DPBS to form a 2% red

blood cell suspension. This suspension was added in a low adsorption 96-well plate (Costar, catalog number 3879), 50 μ L/well, and serially diluted anti-human CD47 antibody samples and the positive control (the anti-CD47 antibody secreted by the clone number CC2C6, Biolegend, catalog number 323102), 50 μ L/well, and mixed well. The plate was placed into a 37° C. incubator and incubated for 2 hours, then observed and pictures were taken.

[0185] As shown in Table 5, several anti-human CD47 clones have strong CD47 binding activity and CD47/SIRP α blocking activity, wherein the in vitro activity of the murine monoclonal antibody secreted by CD47-18-5 clone is shown in FIG. 1. The results of erythrocytes agglutination are shown in FIG. 2. CD47-5-65, CD47-8-21, and CD47-16-5 clones can cause human erythrocytes agglutination, while CD47-5-191, CD47-18-5, and CD47-18-10 clones did not cause erythrocytes agglutination.

TABLE-US-00005

CD47	CD47/SIRP α Test samples binding (EC50, nM)	blocking (IC50, nM)
CD47-5-65	0.245	2.946
CD47-5-191	0.206	1.543
CD47-8-21	0.196	2.443
CD47-16-5	0.267	4.750
CD47-18-5	0.252	1.480
CD47-18-10	0.273	2.097

1.4. Determination of the Sequences of the Monoclonal Antibodies

[0186] The DNA sequences of the antibodies secreted by the clones that have antigen-binding activity, antigen-neutralizing activity but do not agglutinate erythrocytes screened above were determined. Cellular mRNAs were first extracted using the RNAprep Pure kit (Tiangen, DP419). Then the SMART 5' RACE kit (Clontech, catalog number 634849) was used to reversely transcribe the total RNAs extracted from the hybridomas into the first strand of the anti-human CD47 cDNAs, and the primers VHGP: GATTACGCCAAGCTTGCC

AGTGGATAGACAAGCTTGGGTGTCGTTTT (SEQ ID NO: 36) and VLGSP: GATTA CGCCAAG CTTGATGGATCCAGTTGGTGCAGCATCAGC (SEQ ID NO: 37) were designed respectively for PCR amplification of the variable regions. The target bands obtained by PCR amplification were cloned into the linearized pRACE vector (Clontech, catalog number: 634859) in an in-fusion way, and single clones were picked up for DNA sequencing.

Example 2. Preparation of the Chimeric Anti-Human CD47 Monoclonal Antibodies

[0187] The sequence encoding the antibody light chain variable region of CD47-18-5 clone amplified by PCR was set forth in SEQ ID NO: 9, and the sequence encoding the antibody heavy chain variable region was set forth in SEQ ID NO: 11. The complementarity determining region sequences can be obtained after excluding the framework region sequences according to the murine variable region sequences, wherein the amino acid sequences of the three complementarity determining regions LCDR1, LCDR2 and LCDR3 of the light chain are set forth in SEQ ID Nos: 1, 2 and 3 respectively, the amino acid sequences of the three complementary determining regions HCDR1, HCDR2 and HCDR3 of the heavy chain are set forth in SEQ ID Nos: 4, 5 and 7 respectively. The chimeric CD47 monoclonal antibodies constructed for CD47-18-5 clone were named mAb-18-5 and mAb-18-5 N297A respectively, of which the light chains were both obtained by linking the light chain variable region (SEQ ID NO: 9) to the light chain constant region (SEQ ID NO: 13) derived from CD47-18-5 clone. The heavy chain of the mAb-18-5 antibody was obtained by linking the heavy chain variable region (SEQ ID NO: 11) to the heavy chain constant region (SEQ ID NO: 14), and the heavy chain of the mAb-18-5 N297A antibody was obtained by linking the heavy chain variable region (SEQ ID NO: 11) to the heavy chain constant region (SEQ ID NO: 15). The sequence encoding the antibody light chain variable region of CD47-18-10 clone amplified by PCR was set forth in SEQ ID NO: 10, and the sequence encoding the antibody heavy chain variable region was set forth in SEQ ID NO: 12. The complementarity determining region sequences can be obtained after excluding the framework region sequences according to the murine variable region sequences, wherein the amino acid sequences of the three complementarity determining regions LCDR1, LCDR2 and LCDR3 of the light chain are set forth in SEQ ID Nos: 1, 2 and 3 respectively, the amino acid sequences of the three complementary determining regions

HCDR1, HCDR2 and HCDR3 of the heavy chain are set forth in SEQ ID Nos: 4, 6 and 8 respectively. The chimeric CD47 monoclonal antibodies constructed for CD47-18-10 clone were named mAb-18-10 and mAb-18-10 N297A respectively, of which the light chains were both obtained by linking the light chain variable region (SEQ ID NO: 10) to the light chain constant region (SEQ ID NO: 13) derived from CD47-18-10 clone. The heavy chain of the mAb-18-10 antibody was obtained by linking the heavy chain variable region (SEQ ID NO: 12) to the heavy chain constant region (SEQ ID NO: 14), and the heavy chain of the mAb-18-10 N297A antibody was obtained by linking the heavy chain variable region (SEQ ID NO: 12) to the heavy chain constant region (SEQ ID NO: 15). The sequence information of the light chain variable region, light chain constant region, heavy chain variable region, and heavy chain constant region of the chimeric antibodies obtained are shown in Table 6.

TABLE-US-00006 TABLE 6 Information on the chimeric antibody sequences VH comprising the CH comprising the VL comprising the CL comprising the amino acid sequence amino acid sequence amino acid sequence amino acid sequence set forth in the set forth in the set forth in the set forth in the following SEQ ID following SEQ ID following SEQ ID following SEQ ID Antibody NO and composition NO and composition NO and composition NO and composition number thereof thereof thereof thereof mAb-18-5 SEQ ID NO: 11 SEQ ID NO: 14 SEQ ID NO: 9 SEQ ID NO: 13 mAb-18-5 SEQ ID NO: 11 SEQ ID NO: 15 SEQ ID NO: 9 SEQ ID NO: 13 N297A mAb-18-10 SEQ ID NO: 12 SEQ ID NO: 14 SEQ ID NO: 10 SEQ ID NO: 13 mAb-18-10 SEQ ID NO: 12 SEQ ID NO: 15 SEQ ID NO: 10 SEQ ID NO: 13 N297A

[0188] The genes encoding the amino acid sequences of the light and heavy chains of the antibodies above were synthesized by Genewiz Ltd to obtain the nucleotide sequences which were then cloned into the eukaryotic cell expression vector X0GC. Then the expression vector was transfected into the ExpiCHO cell line (ExpiCHO™, Catalog number A29133, invitrogen). The cells were inoculated one day before transfection, then resuspended in fresh ExpiCHO™ Expression Medium (ExpiCHO™ Expression Medium, Catalog number A29100, invitrogen), at the cell density 35×10^5 cells/mL. The cell number was counted on the day of transfection and the cell density was in $(70-200) \times 10^5$ cells/mL range, the viability should be higher than 95%. The cells were diluted with pre-warmed ExpiCHO™ expression medium to the final density 60×10^5 cells/mL. The plasmids and the transfection reagent ExpiFectamine™ CHO reagent (catalog number A29131, invitrogen) were diluted with OptiPRO™ medium (catalog number 12309, invitrogen) according to the transfection volume, and the final concentration of the plasmids was 0.5 µg/mL. The diluted transfection reagent was gently added to the plasmids, let it stand for 1-5 min, then the plasmid transfection reagent complex was added to the cells, the mixture was placed in the cell culture incubator, shaken at 125 rpm at 37° C., 8% CO₂. On the next day after transfection, ExpiFectamine™ CHO Enhancer (Catalog number A29131, invitrogen) and ExpiCHO™ Feed (Catalog number A29101-02, invitrogen) were supplemented, and the cell culture incubator was adjusted to be shaken at 125 rpm, at 32° C., 5% CO₂. Supernatants of cell cultures on day 10 after transfection were collected by centrifugation.

Example 3. Binding Activity and Kinetic Constant of the Chimeric Anti-Human CD47 Monoclonal Antibodies to Human CD47

[0189] A 96-well high-adsorption ELISA plate was coated with 1 µg/mL of the recombinant human CD47 (Sino Biological Inc.) at 100 µL per well using carbonate buffer solution, pH=9.6 at 4° C. overnight. The plate was washed 5 times with PBST, and then blocked using PBST with 1% BSA, 300 µL/well, and incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The anti-human CD47 antibody samples and the control CC-90002 (a patent sequence from WO2016109415) were added, 100 µL/well, incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The horseradish peroxidase-labeled anti-mouse IgG antibody (Abcam, catalog number Ab7068) diluted 1:10000 in PBST with 1% BSA was then added, 100 µL per well, and incubated at 25° C. for 1 hour. The plate was washed 5 times with PBST. The colorimetric substrate

TMB was added, 100 μL /well, and developed at room temperature for 10 minutes. 1M H.sub.2SO.sub.4 was added, 100 μL /well, to stop the development. The absorbance at 450 nm was read on a microplate reader. As shown in FIG. 3, the chimeric anti-human CD47 monoclonal antibodies mAb-18-5, mAb-18-5 N297A, mAb-18-10, mAb-18-10 N297A have good binding affinity to human CD47, stronger than that of the control CC-90002.

[0190] Biacore X100 instrument (purchased from GE) was used to detect the kinetic constant of the chimeric anti-human CD47 monoclonal antibodies binding to its antigen human CD47. This instrument utilizes optical surface plasmon resonance technology to detect the association and dissociation between the molecules coated on the biochip and the molecules to be tested. The main reagent used is Protein A chip (GE Healthcare, 29-1275-57). The experimental procedure is briefly described as follows: the chimeric anti-human CD47 antibodies were diluted in running buffer (1 $\times$ HBS-EP+10 mM HEPES, 150 mM NaCl, 3 mM EDTA, 0.05% surfactant P20, pH7.4). During the capture-binding phase, the antibodies at a concentration of 2 $\mu\text{g}/\text{mL}$ were injected for 60 seconds at a speed of 10 $\mu\text{L}/\text{min}$. During the binding phase, the antigen CD47 was diluted into multiple concentrations with running buffer, injected for 120 seconds at a speed of 30 $\mu\text{L}/\text{min}$ respectively. During the dissociation phase, running buffer was injected for 600 seconds at a speed of 10 $\mu\text{L}/\text{min}$. The regeneration condition is 10 mM glycinate solution, pH1.5. Association kinetic constants and dissociation kinetic constants were analyzed and calculated by Biacore X100 evaluation software. The association kinetic constants, dissociation kinetic constants and dissociation equilibrium constants of the chimeric anti-human CD47 antibodies were shown in Table 7. The data indicated that the association and dissociation parameters of the chimeric anti-human CD47 mAbs are comparable to or stronger than those of CC-90002.

TABLE-US-00007 TABLE 7 Kinetic constants of the chimeric anti-human CD47 monoclonal antibodies binding to human CD47

Test sample	K.sub.on (1/Ms)	K.sub.off (1/s)	K.sub.D (M)
CC-90002	1.52E+06	1.76E-03	1.16E-09
mAb-18-5	1.19E+06	8.23E-04	6.93E-10
mAb-18-10	9.94E+05	1.01E-03	1.02E-09

Example 4. Species Specificity and Binding Specificity of the Chimeric Anti-Human CD47 Monoclonal Antibodies

[0191] Species specificity of the chimeric anti-human CD47 mAbs was determined by ELISA. A 96-well high-adsorption ELISA plate was coated with 1 $\mu\text{g}/\text{mL}$ of the recombinant human CD47, cynomolgus CD47, rat CD47 and mouse CD47, at 100 μL per well using carbonate buffer solution, pH=9.6 at 4° C. overnight. The plate was washed 5 times with PBST, and then blocked using PBST with 1% BSA, 300 $\mu\text{L}/\text{well}$, and incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The chimeric anti-human CD47 antibody samples serially diluted in PBST with 1% BSA were added, 100 $\mu\text{L}/\text{well}$, incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The horseradish peroxidase-labeled anti-human IgG antibody (Chemicon, catalog number AP309P) diluted 1:10000 in PBST with 1% BSA was then added, 100 μL per well, and incubated at 25° C. for 1 hour. The plate was washed 5 times with PBST. The colorimetric substrate TMB was added, 100 $\mu\text{L}/\text{well}$, and developed at room temperature for 10 minutes. 1M H.sub.2SO.sub.4 was added, 100 $\mu\text{L}/\text{well}$, to stop the development. The absorbance at 450 nm was read on a microplate reader.

[0192] The species specificity of the chimeric anti-human CD47 monoclonal antibodies was determined by flow cytometer FACS. Normal human RBCs internally donated, cynomolgus RBCs, rat RBCs and mouse RBCs were collected, washed once with cold PBS (GIBCO, Catalog number 14190) containing 2% FBS (Hyclone, Catalog number SH30084.03). The RBCs (2 $\times$ 10^{sup.6} cells per tube) were resuspended in 100 μL of cold PBS containing 2% FBS, and the serially diluted anti-human CD47 antibody samples and the control CC-90002 and the isotype control were added therein. Flow tubes were incubated on ice for 30 minutes. The cells were washed twice with PBS containing 2% FBS. Then the pellets were resuspended in 200 μL of cold PBS containing 2% FBS and FITC-labeled anti-human IgG antibody diluted 1:1000 (Beijing Zhongshan Golden Bridge, Catalog number ZF0306), incubated on ice for 30 minutes in the dark. The cells were washed twice

with PBS containing 2% FBS, then resuspended in 500 μ L of cold PBS. The cell suspension obtained was detected and analyzed on a flow cytometer (BD, FACS Calibur), and the fluorescence intensities of the RBCs were read.

[0193] Binding specificity of the chimeric anti-human CD47 mAbs was determined by ELISA. A 96-well high-adsorption ELISA plate was coated with 1 μ g/mL of the recombinant human PD1, CD28, CTLA4, ICOS, BTLA, CD47, PD-L2, CD80, CD86, B7-H2, CD47, SIRP α and SIRP γ (all from ACROBiosystems), at 100 μ L per well using carbonate buffer solution, pH=9.6 at 4° C. overnight. The plate was washed 5 times with PBST, and then blocked using PBST with 1% BSA, 300 μ L/well, and incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The chimeric anti-human CD47 mAb samples serially diluted in PBST with 1% BSA and the control CC-90002 and the isotype control were added, 100 μ L/well, incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The horseradish peroxidase-labeled anti-human IgG antibody (Chemicon, catalog number AP309P) diluted 1:10000 in PBST with 1% BSA was then added, 100 μ L per well, and incubated at 25° C. for 1 hour. The plate was washed 5 times with PBST. The colorimetric substrate TMB was added, 100 μ L/well, and developed at room temperature for 10 minutes. 1M H.sub.2SO.sub.4 was added, 100 μ L/well, to stop the development. The absorbance at 450 nm was read on a microplate reader.

[0194] As shown in FIG. 4A and Table 8, the chimeric anti-human CD47 monoclonal antibodies can bind to human CD47 and cynomolgus CD47 with similar affinities, but do not bind to rat and mouse CD47 and has species specificity. At the same time, as shown in FIG. 4B, the chimeric anti-human CD47 mAbs also have very strong binding specificity, only binding to CD47, not to B7 family members, CD28 family members and SIRP γ .

TABLE-US-00008 TABLE 8 Binding activity of the chimeric anti-human CD47 monoclonal antibodies to human/cynomolgus/rat/mouse RBCs

	Human RBCs	Cynomolgus RBCs	Rat RBCs	Mouse RBCs	Test sample
Isotype control	MFI = 3.65	3.95	3.33	3.21	CC-90002
MFI = 101.2	82.17	3.42	3.22	mAb-18-5	MFI = 127.95
111.56	3.36	3.26			

Example 5. Activity of the Chimeric Anti-Human CD47 Monoclonal Antibodies in Blocking the Binding of CD47 to its Receptor

[0195] A 96-well high-adsorption ELISA plate was coated with 1 μ g/mL of the recombinant human CD47-Fc (ACROBiosystems) at 100 μ L per well using carbonate buffer solution, pH=9.6 at 4° C. overnight. The plate was washed 5 times with PBST, and then blocked using PBST with 1% BSA, 300 μ L/well, and incubated at 25° C. for 1 hour, and then washed 5 times with PBST. The anti-human CD47 antibody samples were added, 50 μ L/well, the biotin-labelled SIRP α (ACROBiosystems) was added at a concentration of 4 nM (a final concentration of 2 nM), 50 μ L/well, incubated at 25° C. for 90 minutes, and then washed 5 times with PBST. The Streptavidin-HRP (BD Pharmingen, catalog number 554066) diluted 1:1000 in PBST with 1% BSA was then added, 100 μ L per well, and incubated at 25° C. for 1 hour. The plate was washed 5 times with PBST. The colorimetric substrate TMB was added, 100 μ L/well, and developed at room temperature for 10 minutes. 1M H.sub.2SO.sub.4 was added, 100 μ L/well, to stop the development. The absorbance at 450 nm was read on a microplate reader.

[0196] As shown in FIG. 5, the chimeric anti-human CD47 monoclonal antibodies have CD47/SIRP α blocking activity similar to that of CC-90002.

Example 6. Phagocytosis of Tumor Cells by Macrophages Mediated by the Chimeric Anti-Human CD47 Monoclonal Antibodies

[0197] Obtaining of the mature human macrophages: The human PBMC cells (Lonza, Catalog number CC-2702) were thawed and collected, re-suspended in serum-free RPMI 1640 medium (GIBCO, catalog number 11875) at a cell density of 5 $\times$ 10⁶/mL and seeded in a cell culture flask, and cultured in a 37° C. carbon dioxide incubator for 90 minutes. Discarding the culture supernatant and suspension cells, and the adherent cells were incubated for 7 days in a complete medium (RPMI 1640 with 10% FBS) with the addition of 25 ng/ml M-CSF (Sino Biological Inc.,

catalog number 11792-HNAN). Then the macrophages were harvested and re-suspended in a complete medium (RPMI 1640 with 10% FBS) containing 25 ng/ml M-CSF and 50 ng/ml IFN- γ (Sino Biological Inc., catalog number 11725-HNAS). This cell suspension was inoculated into a 48-well cell culture plate, with 50,000 cells per well, incubated for 1 day to let the macrophages mature for later use.

[0198] Raji tumor cells were stained according to the instruction of the CFSE kit (Life technology, catalog number C34554). Briefly, CFSE was diluted with PBS to a working concentration of 5 μ M and prewarmed at 37° C., centrifuged at 1000 rpm for 5 minutes to collect Raji cells, Raji cells were resuspended with prewarmed CFSE working solution, and incubated at 37° C. for 15 minutes. The Raji cells were washed once with the complete medium, resuspend in the complete medium, incubated for 30 minutes, washed twice with the complete medium, and resuspended in the complete medium for later use.

[0199] The 48-well plate was washed 3 times with the complete medium. The CFSE-stained Raji cells were pre-incubated with the anti-human CD47 antibody samples to be tested and the control CC-90002 and the isotype control for 15 minutes, then added to the 48-well culture plate, and incubated in a 37° C. carbon dioxide incubator for 2 hours. After the incubation, the 48-well plate was washed 3 times with the complete medium, and wheat germ agglutinin, alexa fluor 555 (Life technologies, catalog number W32464) diluted in the complete medium was added, 10 μ g/ml, and incubated in the dark for 15 minutes. The 48-well plate was further washed 3 times with the complete medium, and fixed with 4% paraformaldehyde for 15 minutes. The 48-well plate was further washed 3 times with the complete medium, and the complete medium was added therein. Photos were taken under a fluorescence microscope, and the number of cells was counted. The calculation method of the phagocytosis index (%) is: the number of phagocytosed Raji cells labeled with green/the number of the existing macrophages labeled with red $\times$ 100.

[0200] As shown in FIG. 6, the chimeric anti-human CD47 mAbs can mediate phagocytosis of Raji tumor cells by macrophages, which is equivalent to or stronger than that of CC-90002.

Example 7. Effect of the Chimeric Anti-Human CD47 Monoclonal Antibodies on Red Blood Cells

[0201] Red blood cells (RBC) highly express CD47, and some anti-CD47 antibodies can cause erythrocytes agglutination. Whether the chimeric anti-human CD47 mAbs cause erythrocytes agglutination was determined. Anticoagulated healthy human whole blood donated internally was resuspended in DPBS, centrifuged, the supernatant was discarded, the red blood cells at the bottom of the tube were collected, washed 3 times, and the red blood cells were resuspended in DPBS to form a 2% red blood cell suspension. This suspension was added in a low adsorption 96-well plate, 50 μ L/well, and serially diluted chimeric anti-human CD47 mAb samples and the positive control (the anti-CD47 antibody secreted by the clone number CC2C6, Biolegend, catalog number 323102), 50 μ L/well, and mixed well. The plate was placed into a 37° C. incubator and incubated for 2 hours, then observed and pictures were taken.

[0202] As shown in FIG. 7, only the positive control caused human erythrocytes agglutination, while mAb-18-5 and mAb-18-10 did not cause erythrocytes agglutination.

Example 8. Efficacy Study of the Chimeric Anti-Human CD47 mAbs

[0203] This example examines the growth inhibitory effect of the chimeric anti-human CD47 mAbs on Raji tumor grafts inoculated in NCG mice. The 6-8-week-old female NCG mice (Nanjing Biomedical Research Institute) were selected as the experimental material. After the mice acclimated to the environment for one week, each mouse was inoculated with 5×10^6 human Raji tumor cells. Once the tumors grow to a volume of about 150 mm³, the mice were divided into different groups according to the tumor volumes, 6 mice per group. The vehicle control group, CC-90002 control group, the chimeric anti-human CD47 mAb-18-5 administration group, the chimeric anti-CD47 mAb-18-5 N297A administration group were set up, with a dosage of 35 nmol/kg, twice a week for 2 consecutive weeks, via an intraperitoneal injection. The tumor volumes were measured twice a week beginning at the day of administration, and the long diameter

a and the short diameter b were measured and the tumor volume calculation formula is: tumor volume (mm³)=(a×b²)/2.

[0204] As shown in FIG. 8, the chimeric anti-human CD47 mAb-18-5 has an anti-tumor activity, significantly inhibited the growth of Raji xenograft tumors inoculated in NCG mice, with its efficacy comparable to that of CC-90002.

Example 9. Preparation of the Humanized Anti-Human CD47 Monoclonal Antibodies

[0205] The humanized anti-human CD47 mAbs were obtained according to the method described by Leung et al., *Molecule Immunol.*, 32:1413-1427 (1995).

[0206] The humanized templates that best match the variable region sequences of the murine antibody were selected from the Germline database, wherein the template for the light chain variable region is IGKV1-17*02 of the sequence set forth in SEQ ID NO: 34; the template for the heavy chain variable region is IGHV1-69*01 of the sequence set forth in SEQ ID NO: 35. The CDR regions of the murine antibody were grafted into the selected humanized templates to replace the CDR regions of the human templates, obtaining the grafted, humanized antibody light chain variable region of the sequence set forth in SEQ ID NO: 25, and the grafted, humanized antibody heavy chain variable region of the sequence set forth in SEQ ID NO: 28. Sites on SEQ ID NO: 25 and SEQ ID NO: 28 were selected to carry out back mutations to obtain the light chain variable region sequences set forth in SEQ ID Nos: 26 and 27, and the heavy chain variable region sequences set forth in SEQ ID Nos: 29-33. The light chain variable region was linked to the light chain constant region (SEQ ID NO: 13) to obtain the corresponding full-length sequences of the light chains respectively, and the heavy chain variable region was linked to the heavy chain constant region (SEQ ID NO: 14) to obtain the corresponding full-length sequences of the heavy chains respectively. Available humanized sequences were obtained through affinity and stability screening, and the sequence information of the light and heavy chain variable regions of the obtained humanized sequences is shown in Table 9.

TABLE-US-00009 TABLE 9 The light and heavy chain variable region sequence information for humanized sequences VL comprising the amino VH comprising the amino acid sequence set forth acid sequence set forth Antibody in the following SEQ ID in the following SEQ ID number NO and composition thereof NO and composition thereof Chimeric 9 11 mAbs Z0 25 28 Z1 29 Z2 30 Z3 31 Z4 32 Z5 33 Z6 26 28 Z7 29 Z8 30 Z9 31 Z10 32 Z11 (also 33 known as BH3008b) Z12 (also 27 33 known as BH3011b)

Example 10. Antigen-Binding Activity of the Humanized Anti-Human CD47 Monoclonal Antibodies

[0207] The humanized CD47 monoclonal antibodies were named BH3008b and BH301Tb, in which the variable regions correspond to Z11 and Z12 respectively, the light chain variable region was linked to the light chain constant region (SEQ ID NO: 13), and the heavy chain variable region was linked to the heavy chain region (SEQ ID NO: 16). Detection of the antigen-binding activity of the humanized anti-human CD47 mAbs was performed as described above. As shown in FIG. 9, the humanized anti-human CD47 monoclonal antibodies BH3008b and BH3011b have good binding affinity to human CD47, which is comparable to that of the control CC-90002.

Example 11. Activity of the Humanized Anti-Human CD47 mAbs in Blocking the Binding of CD47 to its Receptor

[0208] The activity of the humanized anti-human CD47 mAbs of blocking the binding of CD47 to its receptor was detected as described above.

[0209] As shown in FIG. 10, the humanized anti-human CD47 mAbs BH3008b and BH3011b have CD47/SIRPα blocking activity similar to that of CC-90002.

Example 12. Binding Kinetic Constant of the Humanized Anti-Human CD47 Monoclonal Antibodies to Human CD47

[0210] As mentioned above, Biacore X100 instrument was used to detect the binding kinetic constant of the humanized anti-human CD47 monoclonal antibodies to its antigen human CD47.

The association kinetic constants, dissociation kinetic constants and dissociation equilibrium constants of the humanized anti-human CD47 antibodies were shown in Table 10. The data show that the association and dissociation parameters of the humanized anti-human CD47 monoclonal antibodies BH3008b, BH3011b are comparable to those of CC-90002.

TABLE-US-00010

Test sample	K.sub.on (1/Ms)	K.sub.off (1/s)	K.sub.D (M)
CC-90002	1.57E+6	1.35E-3	8.57E-10
BH3008b	9.98E+5	1.17E-3	1.17E-9
BH3011b	1.02E+6	8.52E-4	8.33E-10

Example 13. Effects of the Humanized Anti-Human CD47 Monoclonal Antibodies on Red Blood Cells

[0211] Whether the humanized anti-human CD47 monoclonal antibodies BH3008b, BH3011b and the control CC-90002 cause erythrocytes agglutination was determined as described above, wherein the positive control is the anti-CD47 antibody secreted by the clone number CC2C6, Biolegend, catalog number 323102.

[0212] As shown in FIG. 11, only the positive control caused human erythrocytes agglutination, while CC-90002, BH3008b, and BH3011b did not cause erythrocytes agglutination.

Example 14. Detecting the Purity and Thermal Stability of the Humanized Anti-Human CD47 Monoclonal Antibodies by Size-Exclusion High Performance Liquid Chromatography (SE-HPLC)

[0213] Waters Xbridge BEH200 chromatographic column (catalog number: 186007640) was used, with the mobile phase: 0.1 mol/L NaCl buffer, pH 6.7, the flow rate: 0.8 mL/min, the column temperature: 25° C., the sample cell temperature: 4° C., and the detection wavelength: 280 nm. The sample was diluted to 1 mg/mL with the sample buffer DPBS, and the injection volume is 10 µL. Agilent High-Performance Liquid Chromatography 1260 system workstation was used to process the data of the experimental results, and the proportion of the main peak (i.e. the purity) was calculated through the area normalization method. The purities of the humanized anti-human CD47 monoclonal antibodies BH3008b and BH3011b prepared above were detected with SE-HPLC. To determine the thermal stability of these monoclonal antibodies, the above samples were placed under a high temperature condition of 40° C., samples were taken at the 2nd week and 4th week for SE-HPLC detection to observe the thermal stability, and the results are shown in Table 11 below. All the humanized anti-human CD47 antibodies showed good and considerable stability.

TABLE-US-00011

	0 Week	1 Week	2 Week	4 Week
BH3008b	100	100	99.84	99.04
BH3011b	100	100	99.90	99.01

Example 15. Detection of Charge Isomers of the Humanized Anti-Human CD47 mAbs by Ion Exchange Chromatography (IEX)

[0214] The cation exchange chromatography column MabPac SCX-10, 4 mm×250 mm (Thermo, catalog number: 78655) was used, with 20 mmol/L 2-(N-Morpholino) ethanesulfonic acid (MES) (pH5.6) and 60 mmol/L sodium chloride as the mobile phase A, 10 mmol/L MES (pH5.6) and 300 mmol/L sodium chloride as the mobile phase B, the flow rate: 0.6 mL/min, the column temperature: 25° C., the sample cell temperature: 4° C., the detection wavelength: 280 nm, the sample loading: 40 µl (1 mg/mL), the elution mode: 10-55% linear gradient running for 43 min. The data of the experimental results were processed by Agilent High-Performance Liquid Chromatography 1260 system workstation, and the peak area percentage was calculated by the area normalization method. The humanized anti-human CD47 monoclonal antibodies BH3008b and BH3011b prepared above were detected by IEX. To determine the chemical stability of these monoclonal antibodies, the above samples were placed under a high temperature condition of 40° C., and samples were taken at the 2nd and 4th weeks for IEX detection, and the changes in the ratio of charge isomers were observed. The results are shown in Table 12. The changes of the proportions of the charge isomers of the humanized anti-human CD47 antibodies were all relatively low.

- 4.** The anti-human CD47 antibody or functional fragment thereof according to any one of claims 1-3, wherein the light chain variable region has the amino acid sequence set forth in SEQ ID NO. 9 or 10, and/or the heavy chain variable region has the amino acid sequence set forth in SEQ ID NO. 11 or 12, and/or the amino acid sequences of the light chain constant region and the heavy chain constant region are as set forth in any one of SEQ ID NO. 13 and SEQ ID NOs. 14-16.
- 5.** The anti-human CD47 antibody or antigen-binding fragment thereof according to any one of claims 1-4, wherein the framework regions of the light chain variable region of the anti-human CD47 antibody or antigen-binding fragment thereof include FR-L1, FR-L2, FR-L3 and FR-L4, and the framework regions of the heavy chain variable include FR-H1, FR-H2, FR-H3 and FR-H4, wherein the amino acid sequence of the FR-L1 is set forth in SEQ ID NO. 17; the amino acid sequence of the FR-L2 is set forth in SEQ ID NO. 18, or an amino acid sequence as set forth in SEQ ID NO. 18 obtained by one of the following substitutions or any combination thereof: the 12th amino acid R is substituted with T; the amino acid sequence of the FR-L3 is set forth in SEQ ID NO. 19, or an amino acid sequence as set forth in SEQ ID NO. 19 obtained by one of the following substitutions or any combination thereof: the 31st amino acid Y is substituted with F; the amino acid sequence of the FR-L4 is set forth in SEQ ID NO. 20; the amino acid sequence of the FR-H1 is set forth in SEQ ID NO. 21; the amino acid sequence of the FR-H2 is set forth in SEQ ID NO. 22; the amino acid sequence of the FR-H3 is set forth in SEQ ID NO. 23, or an amino acid sequence as set forth in SEQ ID NO. 23 obtained by one of the following substitutions or any combination thereof: the 8th amino acid E is substituted with T, the 11th amino acid S is substituted with N, the 31st amino acid A is substituted with V; and/or the amino acid sequence of the FR-H4 is set forth in SEQ ID NO. 24.
- 6.** The anti-human CD47 antibody or antigen-binding fragment thereof according to any one of claims 1-5, wherein the light chain variable region has the amino acid sequence as set forth in any one of SEQ ID NOs. 25-27; and/or the heavy chain variable region has the amino acid sequence as set forth in any one of SEQ ID NOs. 28-33.
- 7.** The anti-human CD47 antibody or antigen-binding fragment thereof according to claim 6, wherein the light chain variable region has the amino acid sequence set forth in SEQ ID NO. 26 or 27, and/or the heavy chain variable region has the amino acid sequence set forth in SEQ ID NO. 33.
- 8.** The anti-human CD47 antibody or antigen-binding fragment thereof according to any one of claims 1-7, wherein the antigen-binding fragment is one or more selected from the group consisting of F(ab').sub.2, Fab', Fab, Fd, Fv, scFv, diabody, camelid antibody, CDR and antibody minimal recognition unit (dAb), preferably, the antigen-binding fragment is Fab, F(ab').sub.2 or scFv.
- 9.** An isolated nucleic acid molecule selected from the group consisting of: (1) DNA or RNA encoding the anti-human CD47 antibody or antigen-binding fragment thereof according to any one of claims **1-8**; (2) a nucleic acid that is completely complementary to the nucleic acid defined in (1).
- 10.** An expression vector comprising the nucleic acid molecule according to claim 9 operably linked.
- 11.** A host cell comprising the nucleic acid molecule according to claim 9 or the expression vector according to claim 10.
- 12.** A composition comprising the anti-human CD47 antibody or antigen-binding fragment thereof according to any one of claims 1-8, the nucleic acid molecule according to claim 9, the expression vector according to claim 10 or the host cell according to claim 11, and one or more pharmaceutically acceptable carriers, diluents or excipients.
- 13.** A method of producing the anti-human CD47 antibody or antigen-binding fragment thereof according to any one of claims 1-8, comprising the steps of: culturing the host cells according to claim 9 under culture conditions suitable for the expression of the anti-human CD47 antibody or antigen-binding fragment thereof, and optionally, isolating and purifying the obtained products.

14. Use of the anti-human CD47 antibody or antigen-binding fragment thereof according to any one of claims 1-8, the nucleic acid molecule according to claim 9, the expression vector according to claim 10, or the host cell according to claim 11 in the manufacture of a medicament for the prevention and/or treatment of CD47-mediated diseases or disorders, preferably, the diseases or disorders are tumors; more preferably, the tumors are one or more selected from the group consisting of leukemia, lymphoma, myeloma, brain tumor, head and neck squamous cell carcinoma, non-small cell lung cancer, nasopharyngeal cancer, esophageal cancer, gastric cancer, pancreatic cancer, gallbladder cancer, liver cancer, colorectal cancer, breast cancer, ovarian cancer, cervical cancer, endometrial cancer, uterine sarcoma, prostate cancer, bladder cancer, renal cell carcinoma, and melanoma.
